

How We Get Where We Are

People, Institutions, and Luck in an Unequal World

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Preface

This manuscript grew out of an interest in inequality and the explanations we give for it. People differ in ability, effort, interests, and judgment. Yet recognizing those differences leaves much to understand about the rewards they receive, the setbacks they face, and the opportunities they can pass to others. I wanted to follow how differences become consequential, and why some consequences reach so much further than the activity that first produced them.

The question matters because an account of how someone reached a position shapes how we understand the person and what we think can be changed. A result read only as a measure of talent or effort suggests one explanation. Looking also at access, ownership, obligations, and timing may reveal something else that needs attention. The intention here is to keep individual action in view while examining the circumstances through which it becomes effective.

Following that concern led beyond the distribution of income or wealth. An opportunity must be usable; a service must be provided; a cooperative arrangement must be sustained. Understanding who benefits therefore also means asking how people build productive capacity, respond to interruptions, and alter the rules under which they act. Inequality remains the starting concern, but the inquiry also includes what people can accomplish together and how an unfavorable path might be changed.

The discussion is written for readers willing to look closely at familiar experiences without first becoming specialists in each subject. The cases range across housing, farming, health care, technology, finance, and irrigation. Their value here lies in helping us see connections that are easy to miss when a result is viewed alone. The research supplies evidence and challenges; the examples give us something concrete to follow.

My hope is that the reader comes away better able to ask what made an achievement possible, where an obstacle arose, and what an earlier outcome leaves available for the next decision. We need those explanations both to recognize genuine contributions and to judge which arrangements and possibilities deserve to be preserved or changed.

San Francisco, September 2026

Introduction

Imagine two people opening small businesses with comparable skills, experience, and savings. Both work hard to attract customers. One secures an early contract; the other narrowly misses it. The contract pays for better equipment and provides a reference for the next customer. The other owner spends the same months seeking work while drawing down savings.

Several years later, one business has expanded while the other has closed. The owners' decisions belong in the explanation. So do the opportunities available when those decisions were made. An initial difference in earnings has become a difference in equipment, customer relationships, and the ability to survive a difficult period. The later gap raises a question:

Why can relatively modest differences in individual capabilities coexist with much larger differences in rewards, resources, and influence?

The wealth-distribution literature documents pronounced concentration at the top across countries and periods. [1] Our opening example suggests why comparing personal performance is only a beginning. Skills and usable resources must meet an opportunity; selection and the division of proceeds determine the reward. Retained gains can then support another attempt. An accurate ranking of the suppliers would settle only one step in that history.

An initial allocation can change later participation as well as the resources already received. In a Dutch research-funding program, applicants just above an award threshold accumulated, on average, more than twice as much subsequent funding over eight years as applicants just below it, excluding the initial grant. Part of the difference arose because unsuccessful applicants entered later competitions less often. The award affected the conditions under which later opportunities were pursued, giving the first decision consequences beyond its immediate financial value. [2]

To follow these consequences, we need two connected views. The point-in-time view examines what people can do now, which resources and relationships they can use, and how rules and organizations translate those positions into benefits and burdens.

A skill matters differently depending on who has the equipment, time, finance, and customer access needed to use it. These resources may reinforce one another or compensate for a disadvantage. Even when two owners can offer comparable work, a customer's choice determines who receives a particular order.

That choice can depend on information produced by earlier choices. In an experimental music market, Salganik, Dodds, and Watts gave participants access to the same set of unfamiliar songs while varying information about previous participants' choices. Stronger social influence increased both inequality and unpredictability of success. The songs' appeal still mattered, but the distribution of success also depended on what participants could learn about one another's selections. [3]

Once earlier choices affect what later participants see or can use, the present comparison acquires a history. The *dynamic view* follows that history. The successful business's equipment and customer record are both current resources and consequences of earlier activity. Today's outcome becomes one of tomorrow's conditions; a later snapshot records what these transitions have produced.

The two views answer successive parts of the opening puzzle. At a particular time, combinations, selection, and claims shape rewards. Across time, rewards can alter the resources and opportunities available for later activity. The distribution we observe need not resemble the distribution of any single characteristic entering the process.

Property specifies one way a result remains available. In our business example, an asset bought with earlier earnings can stay under the owner's control. Whether the owner can use it, receive its returns, borrow against it, or transfer it to someone else depends on the rights attached to it. Those rights give an earlier result different possible futures. Someone receiving the asset later could begin with a resource position they did not create, so the relevant history extends beyond the present holder's own activity.

Within these histories, individuality, institutions, and luck become consequential together. The owners' skills, effort, and choices affect what they accomplish; institutions shape what they can claim and retain. The timing of a customer's arrival or an unexpected interruption may lie outside an owner's control. That timing can be luck from the owner's perspective while arising from decisions made elsewhere, and its consequences depend on the resources and obligations the owner faces when it occurs.

An advantage need not persist. Shared equipment, another contract, or temporary support might allow the second business to continue; a successful firm's position might weaken as competitors learn or circumstances change. We need to explain recovery and erosion as well as reinforcement.

The inequality question thus reaches beyond dividing an unchanged stock of resources. People and organizations can develop new productive capacity, make existing benefits more accessible, or lose the ability to sustain them. The chapters follow how differences form, how rules and organizations make them consequential, and how each outcome changes the possibilities for what follows.

Empirical studies provide evidence about particular settings and periods. Simplified examples make individual relationships visible, while a mathematical illustration shows what follows under stated assumptions. An illustration demonstrates a possibility; explaining an observed result requires evidence about the process that produced it.

To begin, however, we must look behind the people and positions entering the first comparison. Our imagined owners already possess skills, savings, expectations, and experience. Where did those differences come from? Before following how differences become consequential, we need to understand how they were formed.

CHAPTER 1

Where Do Consequential Differences Come From?

Two young adults applying for their first full-time jobs provide a simple comparison. One has a stronger qualification, more experience, and a clearer sense of how to approach employers. These differences may matter to the hiring decision. To understand what the application forms record, however, we must look at the experience through which the applicants reached these positions.

Even with similar family incomes, they might have attended different schools, encountered different opportunities, or responded differently to opportunities they shared. One might have had help arranging work experience; the other might have needed to spend that time earning money.

The people entering the hiring comparison have already lived through earlier activity. Useful work experience can be both a present qualification and the consequence of previous access. Understanding individual capabilities therefore includes investigating how they developed, alongside assessing how they matter now.

Where children grow up provides a concrete way to investigate that history. Can exposure to a place change what they later become able to do? And how can we distinguish an effect of the place from differences among the families living there?

1.1 What can a map of opportunity tell us?

The Opportunity Atlas documents adult outcomes by the neighborhoods in which children grew up. Its estimates show substantial variation across childhood locations, including nearby areas and comparisons among children whose parents had similar incomes. The map makes a question visible: why do children from these places reach different adult positions? [4]

The map alone leaves the effect of place entangled with the process by which families arrive there. A family seeking a home might weigh the rent, the journey to work, proximity to relatives, and the schools available. The home it obtains also depends on what is affordable and which landlords will accept its application. Two families with the same income could consequently reach different places for reasons that also matter to their children.

Family employment connections show how that entanglement can arise. Existing connections could influence both where a family settles and the opportunities available to its children. Living in the destination could also open relationships that the family lacked before. Both routes could contribute to later earnings, but only the second attributes the new connections to exposure to the destination.

To separate those routes, we need comparisons that change the opportunity to move without simply comparing families who chose different destinations. A second approach examines differences in how much of childhood is spent in a destination. The studies that follow use these distinct comparisons to investigate what place contributes.

1.2 What happens when the opportunity to move changes?

The Moving to Opportunity (MTO) demonstration created such a comparison. It recruited low-income families living in distressed public or assisted housing in five U.S. cities. One randomly assigned group was offered housing vouchers and assistance intended to support a move to a lower-poverty neighborhood; comparison groups received different assistance or no new voucher through the demonstration. The random assignment concerned the offer, not a compulsory relocation. Some families offered the opportunity moved and others did not. [5, 6]

Comparing randomly assigned offers provides a firmer basis for studying what the opportunity changed than comparing families who independently chose different neighborhoods. Estimating the effects of moves induced by the offer requires a further step; the movers themselves were not randomly selected. [5]

The offer-based comparison also requires us to specify whose outcome and which stage of life we are examining. Earlier evaluations found improvements in neighborhood safety and aspects of health, but little detectable improvement in adults' employment or earnings. Youth outcomes also differed across groups. [6–8]

Later work by Chetty, Hendren, and Katz followed children into adulthood using administrative records. Children who moved to lower-poverty neighborhoods when young—below age thirteen in the study's main comparison—experienced gains in college attendance and adult earnings. Older children did not show comparable benefits, and the estimated gains declined with age at move. The authors considered both the duration of childhood exposure and the disruption of moving in interpreting this pattern. [5]

The earlier adult results and later childhood findings concern different people, stages of life, and outcomes. They can coexist because moving changes an adult's present surroundings while also changing part of the environment in which a child develops. The age pattern gives us a reason to examine the duration of that experience more closely.

1.3 Why does the age at move matter?

To investigate whether duration mattered beyond the MTO demonstration, Chetty and Hendren studied more than seven million families moving across U.S. counties and commuting zones. Children's later outcomes moved toward those associated with the destination in proportion to how much of childhood they spent there. Their related work estimated differences in childhood exposure effects across counties. [9, 10]

These were observational studies of family moves, not randomized assignments. The researchers examined alternative explanations using comparisons among siblings, moves associated with displacement, and other tests. Their causal interpretation depends on how successfully the designs separate exposure from other differences among families and their histories. [9, 10]

A family moving with children aged eight and sixteen illustrates why exposure differs. Both children enter the same home on the same day. Yet the younger child will spend more of childhood there, while the older child brings a longer history of schooling and relationships into the move. The same event in the family's record creates different developmental experiences.

A new environment might provide opportunities that take years to use. Moving might also interrupt friendships or schooling, with different consequences at different ages. These remain possible explanations of the age pattern rather than separately established mechanisms. [5] The evidence supports a causal contribution of exposure to place, with duration helping explain its consequences, while leaving the particular features doing that work to be investigated. [5, 9, 10]

1.4 What about the neighborhood does the work?

Identifying a contribution of place does not yet identify which features of the place produced it. The MTO offer changed a residential situation rather than assigning schools, safety, relationships, stress, and other influences separately. The mover studies likewise identify consequences of exposure to places, leaving the individual contributions of those features open. [5, 6, 9, 10]

A move can alter a child's school journey, classmates, access to outdoor space, and contact with nearby relatives. These influences may depend on one another. A shorter journey could make an after-school activity practical; losing help from a relative could make another activity harder.

Because several features can change together, evidence about a move cannot automatically tell us what a school improvement or support for a local organization would accomplish on its own. Those are different actions affecting different parts of the child's environment. A proposal to reproduce the benefit must identify which parts it changes and whether the other supports remain available.

In the after-school example, access depends jointly on distance, family arrangements, and the activity's availability. The relevant question is whether this child can use them together. That requirement begins even before exposure to the new neighborhood: the family first has to obtain a home there.

1.5 Can the family use the opportunity it receives?

A housing voucher can help meet the cost of a move, but the family also needs to locate a suitable home, complete an application, and secure a landlord's agreement. Assistance with rent and access to an actual tenancy address different parts of the same task.

The Creating Moves to Opportunity experiment in Seattle and King County investigated these barriers among families receiving housing vouchers. It tested support that included information, financial assistance, and individualized help with housing search and landlord engagement. The intervention substantially increased moves to neighborhoods with higher rates of upward mobility. A further experimental comparison found that customized search assistance was especially important to the result. [11]

The measured outcome here was the move, not those children's eventual adult earnings. The earlier neighborhood studies explain why the destination could matter; the search-assistance experiment identifies a way to make that destination more accessible. [5, 9–11] The evidence connects two successive steps without treating a successful move as a measured long-term earnings gain.

Those steps place policy implementation within the formation of individual differences. Formal assistance, a housing search, the landlord's response, the realized move, and subsequent exposure link the family's circumstances to the child's experience. An obstacle before the move can prevent the later experience from taking place.

This also changes how we interpret the family's decision. Its destination can reflect restricted options as well as preferences, including whether a landlord would accept its application. The residential experience develops through the choices the family makes and the opportunities that housing rules and organizations make usable.

1.6 How do circumstances and choices interact?

Making an opportunity usable still leaves someone to act on it. The applicant who gained useful work experience had to do the work; access made that effort possible. Interests, abilities, judgment, and responses to shared experiences can also differ. These remain part of the explanation alongside the contribution of neighborhood exposure established by the research. [5, 9, 10]

A skill can consequently be useful now and have developed through earlier access; an expectation can guide a choice now and reflect previous experience. Neither the present choice nor its history has to displace the other from the explanation. Together, they show how people act within circumstances that their earlier activities may also have helped to change.

Some of those circumstances precede any choice the person could make. For a child, parents' resources, the initial home, and the timing of a move can count as luck, while remaining connected to decisions and constraints elsewhere. The neighborhood evidence shows that some early circumstances can be altered, with later consequences that depend on timing. [5, 9, 10] An unchosen starting position can therefore matter without determining the whole path.

1.7 What does a later classification record?

An employer's assessment can identify which applicant is better prepared at the hiring date. Explaining how that preparation developed is another task; deciding how to reward it is another. The assessment can inform the hiring decision while leaving room to ask how the less-prepared applicant might acquire the missing experience.

The hiring example connects the chapter to the point-in-time puzzle. A cross-sectional comparison can take education, skills, health, or connections as current characteristics; the dynamic account asks how earlier opportunities and experiences helped form them. What looks like an input to the present hiring decision may be an outcome of an earlier allocation.

A mobility program could change a later distribution, but its population effect depends on who participates, what happens to others, and how later institutions translate changed opportunities into rewards. Gains among participants identify one part of that process, not its effect on inequality as a whole.

The next part concerns what people can do with capabilities and resources formed through these histories. A skill may qualify someone for one opportunity and be excluded from another. A resource may be usable for a season or remain under someone's control for years; a gain may be consumed, retained, or transferred. To follow these consequences, we must examine the rules governing use, reward, and continued control.

CHAPTER 2

How Do Rules and Property Make Differences Matter?

China's Household Responsibility System (HRS) expanded household control over agricultural production while farmland remained collectively owned. Beginning in the late 1970s, households gained greater discretion over farming and a stronger claim on the returns from their work. [12, 13] How could the consequences of working the land change so substantially while the ownership label remained the same?

The distinction concerns what households could decide, what they stood to gain, and how long they expected their claims to last. Land has productive possibilities; the rights attached to it determine who may use those possibilities and receive the returns. Examining those rights connects the present use of a resource with what its holder can carry into the future.

2.1 What does holding a resource allow someone to do?

Consider first the duration of a use right. A one-season cultivation right creates a different investment horizon from a claim expected to last for many years. Both allow a household to use the land, but only the longer claim may let it collect returns from an improvement that takes several seasons to repay its cost.

Duration is separate from the actions the right permits. Permission to let another farmer use the field in return for payment adds an option without adding land. The ability to exclude an unauthorized user adds another form of control. Use, management, income, exclusion, and transfer can belong to different people and last for different periods. Deakin and Meng's analysis of the Household Responsibility System emphasizes this division of rights among households and collective bodies. [14]

An ownership label therefore leaves much of the household's practical position unspecified. A long-term claim, a transferable claim, and a claim on income offer different possibilities even when they concern the same field. To understand economic control, we need to know which rights the household holds and whether it can exercise them.

The distinction extends to the business discussed in the Introduction. A worker may operate a machine while its owner receives income from its use. A long-term lease may support one investment decision and a revocable permission another. What one person controls helps determine what another must obtain permission to use, pay for, or go without. Property organizes relationships among people around the use of resources.

2.2 What changed under the Household Responsibility System?

Before the reforms, agricultural production was organized substantially through collective structures associated with the people's communes. Production decisions and remuneration were collective to a significant degree, weakening the connection between an individual household's additional effort and the return that household could keep. [12, 13]

Under the Household Responsibility System, households became the principal units responsible for farming their allocated land. They gained greater discretion over production and retained more of the output or income remaining after required obligations were met. Collective ownership persisted alongside these changes in household control and claims on proceeds. [12, 13, 15]

What remains after obligations and costs are met is often called the *residual return*. A household's interest in producing another unit depends partly on how much of that return it can keep. The reform altered the connection between household decisions, additional work, and reward.

Whether the stronger claim actually changed production is an empirical question. McMillan, Whalley, and Zhu examined the contributions of price changes and changed individual incentives to agricultural productivity growth after 1978. They attributed a substantial part of the increase to changes in the payment system. Lin's province-level analysis also found a large contribution from decollectivization, while distinguishing it from price adjustments and other reforms. [12, 13]

The studies estimate contributions within a historical reform, rather than a randomized experiment holding other influences fixed. Their estimates differ, but both support an important role for the changed relation between household activity and returns. [12, 13]

2.3 How can different rules change the use of the same land?

To see how the changed payment system could affect production, separate decision authority from claims on returns. Authority determines which production decisions the household can make; its claim on the yield determines what it stands to gain. A household holding both rights can respond differently from one permitted to choose a method but unable to retain its additional yield.

For a household working the same plot, greater discretion could make it possible to carry out an additional task at a useful time. Keeping more of the resulting harvest could make that task worthwhile. The household may use its knowledge differently or organize its work more carefully while the physical plot remains unchanged.

A farmer's skill and effort operate through these opportunities. The rules affect how skill can be used and what its use yields; the farmer still has to choose and carry out the work. Individual action and institutional conditions belong in the same explanation.

The same stronger connection to returns can also change exposure to loss. A household that keeps the benefit of an unusually good harvest may bear more of the cost of a poor one. Whether it can sustain its chosen activity then depends partly on the reserves or support available when the harvest disappoints.

Rules can change what a resource makes possible without changing the amount of that resource.

They allocate decision authority and the resulting benefits and losses. Explaining the household's return requires connecting its land and effort to those claims and sources of support. It also leaves a historical question: how did households acquire permission to operate under the new arrangement?

2.4 How did a different arrangement become an accepted rule?

Deakin and Meng trace the Household Responsibility System's development through experimentation, local protection, and wider political acceptance. The 1978 agreement among households in Xiaogang became a prominent episode, but responsibility arrangements also developed elsewhere, including Anhui

and Sichuan. Local practice, provincial support, and central authorization played different roles in the reform's spread. [14]

The reach of that authorization changed as adoption spread. The World Bank and China's Development Research Center report that by the end of 1983 the new system covered the large majority of rural households. [15] A practice initially dependent on particular local support had become a much wider arrangement.

For a household considering the practice, wider recognition changed the risks of acting. Uncertain local tolerance offers different security from support by broader authority. Recognition can alter expectations about continued use and about how other organizations will respond, making a production option more dependable.

The historical account traces how local results gained influence through interpretation, protection, and endorsement. [14] It explains the acquisition of rights whose subsequent operation the productivity studies examine. Gaining those rights made a stronger connection between work and reward possible; it did not resolve every decision about investing in the resources on which production depended.

2.5 Does a better incentive for effort solve the investment problem?

A household might find it worthwhile to tend its current crop more carefully while being reluctant to finance an improvement that benefits several farms over many years. The first decision concerns a return the household can obtain directly. The second concerns how benefits are shared and who will hold the relevant rights when they arrive.

Choe develops this distinction in examining China's rural reform. His analysis argues that household residual claims strengthened work incentives but could leave inadequate incentives for investments whose benefits extended beyond the investing household. In Choe's account, the productive gain coexisted with a further problem of investment and coordination. [16]

A shared drainage channel illustrates the difficulty. One household could pay for an improvement from which neighboring fields would also benefit. Even when the total benefit exceeds the cost, the paying household might recover too little to make paying alone worthwhile. A stronger claim on its own harvest leaves the financing of the shared channel to be resolved.

Even when benefits fall mainly on the investing household, the timing of their receipt can create a problem. A household expecting to lose a plot before an improvement pays for itself has less reason to finance it. Someone seeking to revise the allocation of land may value flexibility instead. Rules governing duration and reallocation must address both the investment horizon and the reasons for reallocating the resource.

Other investments require a larger operating scale than separate plots provide. Individually organized production may still benefit from joint investment, shared services, or combining land use. These options require decisions about contributions, control, and shared benefits—the very questions that a stronger claim on one's own harvest leaves open.

The HRS case thus links household effort to household returns while exposing separate dependencies between investment costs and shared benefits, and between present control and future claims.

Strengthening the incentive to work cannot by itself supply coordination or a secure investment horizon.

Private and collective decisions can perform different, complementary tasks within the same productive activity.

2.6 Which consequences can be carried forward?

The investment horizon and the ability to reorganize land use remained subjects of later reform. The 2022 World Bank report describes extensions of contract duration and changes separating ownership, contracting, and management rights and enabling land-use transfers. Collective ownership persisted as these other rights developed. [15]

Duration, transfer, and retention each carry consequences forward in a different way. A durable claim provides a longer horizon for planning. A transferable right permits exchange. Retained proceeds remain available for later use. These possibilities depend on the rights attached to the particular resource.

Equipment bought with a household's retained earnings makes the connection visible. The harvest has ended, but part of its result is available for the next season. Where the household also has the right to transfer that equipment to a child, the gain reaches someone who did not produce it. The equipment example concerns rights of retention and transfer; households' rights over collectively owned land remain those specified by the agricultural arrangements.

This extension from current earnings to retained assets matters to the inequality question. Two people with similar current earnings can hold different stocks of assets or claims on future income, reflecting earlier activity or transfers. Their present resource positions cannot be explained by this year's skill and effort alone; we must also follow what could be retained and passed on.

Retention makes persistence possible, not assured. Assets can deteriorate or lose value, and claims can be limited, contested, or revised. A right can shape the return to an action today and the resources available for another action tomorrow, while losses or changes in that right can interrupt the connection.

2.7 From rights to realized benefits

The land remained collectively owned while household decision authority and claims on returns changed. Those changes helped alter productive behavior. [12, 13] Longer-run investment, coordination, and security remained consequential alongside the initial gain. [15, 16] The case shows why the consequences of a resource depend on the decisions and claims organized around it.

For the present inequality question, the key relationship runs from rights over resources to the benefits received. Equally skilled farmers on identical plots could receive different returns under different rules; a sharing arrangement could instead moderate differences in returns when production differs. Higher production, security of control, and the division of benefits and burdens would each supply a distinct consideration when we evaluate the arrangement.

Before evaluating those benefits, we must establish whether a claim becomes effective in practice. A household may hold a use right while facing a dispute over access; a person may qualify for a public program while struggling to obtain its benefit. Rights specify a claim, while recognition, administration, and service provision determine how it can be exercised. Chapter 3 follows that path from formal entitlement to something a person can actually use.

CHAPTER 3

Who Gets What—and How?

Two people who qualify for the same health-insurance program can follow different paths. One enrolls, remains covered, and obtains the appointments needed. The other encounters a problem completing renewal or cannot find a listed provider accepting new patients. The benefits each can use depend on what happens between eligibility and care.

Medicaid makes these steps visible through enrollment, renewal, provider networks, and appointments. The benefits do not all arrive at the end of the path: insurance can provide financial protection before treatment is needed and without a measurable improvement in health. [17, 18] We therefore need to follow both the benefits received along the way and the actions required to keep them available.

3.1 What has been allocated when someone qualifies?

An eligibility rule specifies who qualifies. Providing the benefit requires further action: the agency must establish eligibility, the person must become enrolled, and coverage must remain in effect. When treatment is needed, the person must find a suitable provider and obtain an appointment. A university's selection of applicants visibly allocates places; a public insurance program allocates benefits through a sequence of administrative and service decisions.

Each administrative step also allocates work. Requiring applicants to locate documents and respond to notices places tasks on them; verifying information within an agency places more of the work on the administration. That division affects whether protection begins or continues, so the demands made on beneficiaries are part of the allocation rather than merely paperwork surrounding it.

What those steps secure is initially insurance coverage. Coverage redistributes financial exposure to illness and can protect against a possible bill before treatment is needed. The Oregon evidence below found reduced exposure to medical financial liabilities as well as increased service use. [17, 18] Realized allocation can therefore include continuing protection, services received, and the burdens of obtaining them, with health consequences requiring a further assessment.

3.2 What does obtaining insurance change?

We first need evidence about what obtaining coverage changes. People who enroll in Medicaid can differ from those who do not in health needs, knowledge, and ability to navigate the program. A direct comparison of their later outcomes would mix the effects of insurance with those pre-existing differences. The Oregon Health Insurance Experiment supplied a stronger comparison.

In 2008, Oregon used a lottery to select people from a waiting list for the opportunity to apply to a limited Medicaid program for low-income uninsured adults. Selection increased the probability of insurance coverage by about 25 percentage points. Researchers used this randomized variation to estimate the effects of coverage induced by the lottery. The assignment concerned the opportunity to apply, not automatic enrollment; the findings concern adults who had sought access to this particular program. [17]

During the first year, coverage increased medical use and reduced financial strain, including out-of-pocket spending and the risk of unpaid medical bills being sent to collection. Approximately two years after the lottery, a follow-up examination found greater use of services, including preventive care, and increased diabetes diagnosis and use of diabetes medication. It also found fewer positive screenings for depression and substantial protection against catastrophic out-of-pocket expenditure. [17, 18]

Greater use of services did not translate into a detectable gain on every health measure during that follow-up. The study found no statistically significant improvements in blood pressure, cholesterol, or glycated hemoglobin, a measure used to assess blood-glucose control. This result concerns the selected measures over the observation period; it does not establish that every health effect was zero. [18]

The combined findings distinguish financial protection, service use, and health. Reduced financial strain is a benefit in its own right, and fewer positive depression screens identify a mental-health gain. More service use records greater utilization, whose health consequences require their own assessment. Timing also affects the comparison: financial exposure may change promptly, while some health effects may require different treatment or longer observation. A claim about such later gains would need evidence from that later period.

3.3 Who must act to keep coverage in place?

The benefits of coverage depend on its continuation, which renewal brings back into an administrative decision. During the COVID-19 emergency, a continuous-enrollment provision largely prevented Medicaid disenrollment. States resumed regular disenrollments in 2023 as they redetermined eligibility. This return to ordinary renewal processes, often called the *unwinding*, generated records of who remained covered and why coverage ended. [19, 20]

Those records distinguish a failure to complete renewal from a determination that someone no longer qualifies. KFF's archived summary, updated September 12, 2024, recorded more than 25 million disenrollments and more than 56 million renewals. Among disenrollments with reported reasons, 69 percent were classified as procedural rather than as determinations of ineligibility. [20]¹

A procedural termination leaves the person's substantive eligibility unresolved in that record. [19, 20] To understand what happened, we must therefore examine the work required to complete renewal, not assume that everyone whose coverage ended had ceased to qualify.

In a notice-and-deadline process, continuation depends partly on the beneficiary's response. Coverage could end after an incomplete response even when the underlying income has not changed. If the agency can instead verify the necessary information from records it already holds, the same eligibility decision may be completed without requiring that additional step from the beneficiary.

Verification using available information without a separate beneficiary response is called *ex parte* renewal. A 2025 study examined federal assistance to improve such renewals in four states. Relative to other states, *ex parte* renewals increased by 21.6 percentage points, overall renewals by 7.7 points, and

¹ The KFF summary combines historical renewal records covering different reporting periods across states, with varying detail. The counts are not current Medicaid enrollment totals. The 69 percent figure applies only to disenrollments with a reported reason. [20]

procedural denials declined by 8.3 points. The study compared state administrative data; the four states were not randomly assigned. [21]

The administrative change is consequential because it moves part of the verification task from the beneficiary to the agency. The study associates that change with more continuing coverage. [21] Verification remains necessary, but how it is organized determines which information is used and whose action continuation requires. Keeping coverage in place, however, still leaves the availability of care to be examined.

3.4 Can a covered person obtain an appointment?

Once a person is covered, the plan establishes the financial terms of seeking care. Obtaining care also requires a suitable provider with an available appointment. Network listings and appointment requests examine different stages between those financial terms and a usable service.

Wallace, Lollo, and Ndumele's 2020 study, using 2017 data from five states, found that the average Medicaid managed-care network included more office-based physicians than the average exchange-plan network in the same geographic markets. Participation estimates were higher when based on directories than when based on physicians' survey answers. These findings concern network inclusion; an appointment request tests whether an included provider can actually see a new patient. [22]

An investigation by the U.S. Department of Health and Human Services Office of Inspector General (OIG) tested that next stage by calling listed primary-care and specialist providers to request appointments. Published in 2014, it used calls made in 2013 to providers listed by Medicaid managed-care plans. [23]²

Projecting the results to the represented provider population, OIG estimated that slightly more than half could not offer an appointment: 35 percent could not be found at the listed location, 8 percent said they did not participate in the plan, and another 8 percent were not accepting new patients. Among those offering appointments, the median wait was about two weeks; more than one-quarter had waits exceeding a month. [23]

The different reasons for an unsuccessful request locate different obstacles. Correcting an address could prevent a wasted call; it would not open a closed patient list or bring a nonparticipating practice into the plan. The percentages are provider-level estimates for the study's sampling frame and period, not beneficiary-level measures or a current national access rate.

A later OIG evaluation examined the information problem in maternal-health directories for plans associated with three large managed-care parent companies in five states. Published in 2026, it found listings for providers who were not in network, missing listings for some in-network providers, and inaccurate contact details. The evaluation concerns information available to someone seeking care, rather than subsequent treatment or maternal-health outcomes. [24]

Accurate information still leaves the provider's decision about accepting a patient to be explained. Using simulated-patient calls across ten states, Candon and colleagues found that changes in Medicaid primary-

² OIG drew a stratified random sample of 1,800 listed primary-care and specialist providers. After exclusions and nonresponse, the analysis used 1,092 provider observations from calls made in 2013. Its reported percentages project to the provider population represented by that sampling frame. [23]

care fees were associated with changes in appointment availability for new Medicaid patients during the period examined. [25] This observational comparison adds payment to the possible influences on availability. Network breadth, directory accuracy, and willingness to offer appointments thus concern distinct connections between coverage and care.

3.5 Where does an organization change the outcome?

The beneficiary encounters the combined operation of several organizations. In the arrangements examined here, agencies determine and renew eligibility, managed-care plans organize networks, and provider practices decide whether and when appointments are available. [19, 21–25]

Each organization acts on a different part of the path. Automatic renewal can preserve coverage while an unavailable appointment prevents care. A broad provider network can offer options while an incomplete renewal interrupts insurance. Locating the obstacle helps identify a response: better information systems could support verification, a corrected directory could prevent wasted calls, and additional clinical staff could expand appointment capacity.

Which step the organization requires also determines which personal resources become consequential. Two equally eligible people receiving the same notice may have different time, internet access, or help resolving an error. The same instruction can demand different efforts and lead to different coverage outcomes. Revising the task can alter the consequences of unequal support, while leaving the substantive eligibility rule unchanged.

3.6 What does “who gets Medicaid” leave out?

Because different steps can succeed or fail, an enrollment count brings together people with different experiences. A person may remain insured while waiting for care, obtain treatment and avoid a large bill, or lose coverage during renewal without a determination of substantive ineligibility. To follow what each receives, we need measures of continuity, protection, access, and health rather than treating enrollment as a complete account of the benefit.

The studies make those distinctions observable. Oregon’s randomized insurance opportunity supports estimates of the effects of induced coverage on service use, financial protection, and measured health. [17, 18] Administrative records distinguish renewals from recorded reasons for termination, while comparisons of renewal procedures investigate continuity. [19–21] Directory audits test listed information, and appointment studies examine whether providers offer care. [22–25]

These studies illuminate successive parts of the path without observing one common set of beneficiaries through every stage. They concern different populations, states, and years; an appointment audit leaves open whether someone later obtains care elsewhere, just as a procedural-termination record leaves substantive eligibility unresolved. Their percentages cannot be multiplied or added into a single national probability of reaching care. Their combined value is to identify distinct processes that a complete account would need to follow.

The connection to unequal outcomes is now concrete: **people meeting the same eligibility rule can encounter different administrative demands, provider options, and practical support.** The

distribution of benefits reflects how those circumstances operate together. Classification identifies who qualifies; the work of agencies, plans, and practices helps determine what follows.

3.7 What remains a question of judgment?

The practical differences just identified turn equal treatment into more than a question of applying the same eligibility rule. Giving eligible people a reasonably comparable ability to exercise their entitlement may require attention to the tasks they can complete. Automatic verification and a beneficiary-response requirement can distribute work differently while serving the same eligibility decision. Choosing between them requires reasons about administrative cost, reliable verification, privacy, and responsibility for supplying information.

Oregon's evidence of financial protection supplies a reason to value coverage. [17, 18] Assessing its priority also requires considering other uses of resources. Identifying an access barrier gives a problem to address; a proposed remedy must be assessed for its ability to remove that barrier and for the costs and burdens it introduces.

Explanation and evaluation meet at the proposed remedy. We establish what people receive, what they must do, and where the process affects their opportunities; we then assess whether changing that process removes an objectionable burden and what new costs it brings. Chapter 7 develops these reasons for judgment across the cases. One distinction needs further development first: whether the obstacle prevents access to a service already available or reflects a shortage of the service itself.

3.8 From realized access to the ability to provide

Correcting a directory can help someone reach care already available. Providing additional care may instead require more clinicians, facilities, or better organization. The beneficiary's path leads us to both tasks, but they call for different actions: connecting people to an existing service and developing the ability to supply more of it.

The ability to provide care also changes through time. People acquire skills, organizations learn or lose the capacity to coordinate, and equipment and technology expand or restrict what can be accomplished. The benefits available for allocation change with them. We next examine how productive capacity is built, maintained, and reorganized, and how those changes alter participation in its benefits.

CHAPTER 4

How Do Social Systems Build— and Reallocate—Capacity?

Better scheduling could help a health service redistribute appointments and reduce unused clinical time. Training additional staff could improve future provision while taking time away from today's service. The same action can affect both who obtain care and how much care the service can provide.

We now examine changes in the amount, quality, and organization of what people can produce, provide, and sustain. Industrial robots offer another way to follow the connection introduced by the clinic: a machine can expand productive possibilities while changing the tasks, organizations, and opportunities through which people participate.

4.1 Capacity is an ability, not simply a quantity of resources

We use *capacity* to mean the ability of people and their organizations to carry out activities and sustain results under particular conditions. The relevant ability might be producing food, delivering treatment, maintaining infrastructure, developing knowledge, or coordinating a response to disruption.

Equipment becomes usable through staff, knowledge, and working arrangements. An organization with sophisticated equipment but too few people able to operate it faces a different constraint from one that needs a better way to coordinate its existing resources. Understanding capacity requires examining how those resources work together.

Because capacity includes the ability to sustain an activity, observed output is only part of the evidence about it. Output records what was accomplished during an interval. A service might temporarily produce more by exhausting its staff or postponing maintenance, while another produces less during investment that improves what it can subsequently deliver. The same comparison of current output could therefore conceal different prospects for continued provision.

Those prospects depend on how today's resources and decisions support tomorrow's activity. People bring abilities formed through earlier experience; authority and working arrangements shape how they can use them. Allocating training, finance, or equipment changes the combination an organization can put to work. A new technology is one such change, whose productive potential must be made usable within that combination.

4.2 A new technology changes possibilities before it changes outcomes

For a manufacturer considering a robot, performing a task is one part of adoption. The machine must also fit into a functioning production process. Workers may lose a task, take on a revised role, or find their skills more useful in the new combination. Following adoption means examining the machine's performance, the organization of production, and the resulting opportunities for people.

Acemoglu and Restrepo's task-based analysis distinguishes two processes. Automation transfers some tasks previously performed by workers to capital, potentially raising productivity while reducing demand for labor in those tasks. Creating new tasks in which people have a comparative advantage can increase demand for their work. The authors call these processes *displacement* and *reinstatement*. Their balance depends on the tasks actually displaced and created; sufficient new tasks do not follow automatically from automation. [26]

Evidence about robot adoption shows why productivity and employment must be followed separately. Graetz and Michaels examined industries in seventeen countries between 1993 and 2007. Their estimates indicate that increased robot use contributed to growth in output per hour worked, or labor productivity; increased output relative to combined measured inputs, or total factor productivity; and lowered output prices. [27, 28] They did not find a statistically significant reduction in total employment in their analysis, but they did find a reduction in the employment share of low-skilled workers. [27]

Productive gains can therefore coexist with uneven employment effects. Even a total employment figure that changes little can conceal shifts among workers or organizations. To understand who benefits from increased capacity, we must locate the gains and follow how adoption changes demand for work inside and outside the adopting firm.

4.3 An expanding firm is not the same as an expanding opportunity for everyone

Acemoglu, Lelarge, and Restrepo's study of French manufacturing firms shows why the location of a gain matters. Robot adopters increased productivity, value added, and overall employment, while the labor share and the share of production workers in their employment declined. The authors also find that adopters expanded at competitors' expense, producing an overall negative association between robot adoption and employment in their wider analysis. [29]

A manufacturer that becomes more productive might lower prices, attract customers, and expand production. A sufficiently large expansion could increase its employment even while fewer workers are needed for each unit of output. Where those customers previously bought from competitors, the competitors could contract, reduce hiring, or close. The adopting firm's employment record captures one part of this wider adjustment.

The example separates an increase in what resources can produce from a shift in where production occurs. A technology can do both at once: improve the adopter's efficiency and draw activity away from competitors. The adopting firm's success therefore supplies only part of the evidence about the wider gain and its distribution.

Following those competitive responses also brings non-adopters into the explanation. Particular organizations adopt, delay, experiment, or decline; their choices affect workers, suppliers, and communities connected to other firms. A technical possibility becomes operating capacity through organizations, and its employment consequences extend through the relationships among them.

4.4 Existing workers and people seeking entry may face different changes

To observe those wider consequences, Acemoglu and Restrepo studied U.S. local labor markets rather than only adopting firms. They found negative effects of industrial-robot exposure on employment and

wages. [30] That broader unit of comparison includes people whose employers did not themselves adopt robots.

The German evidence examined by Dauth and colleagues shows a different adjustment pattern. They find manufacturing displacement associated with robot exposure, offset by additional service employment. Established workers experienced greater employment stability within firms, often by taking on new tasks in their original plants. Much of the adjustment burden fell instead on young people entering the labor market. [31]

Fewer opportunities can appear as reduced hiring of entrants, while existing workers remain employed in altered roles. The German study makes these routes visible. [31] Following employment therefore requires asking who remains inside an adapting organization, who changes employer or occupation, and who finds that a previous route into work has narrowed.

The contrast is informative about possible forms of adjustment, not a controlled test of national institutions. The studies examine different settings, units, and processes, and their findings do not isolate which institutional differences account for the results. [27, 29–31] They show why an account of adjustment must follow the opportunities available to particular workers rather than infer their experience from one firm's output or one aggregate employment total.

Reorganizing production consequently reorganizes the routes through which people participate and receive returns. When fewer days of work are required for an existing task, another productive use of that time has to become available before reduced labor requirements can become a new earning opportunity.

4.5 Where can the released labor go?

The need for another productive use of released labor is not confined to automation. In China's Household Responsibility System, changed rights and incentives helped raise agricultural productivity. McMillan, Whalley, and Zhu attributed a substantial part of the post-1978 increase they examined to changes in the payment system. [12] The wider transformation also involved price reform, modern inputs, infrastructure, industrial development, and opportunities outside agriculture. [15] We can now follow the productive gain beyond the farming decisions considered in Chapter 2.

A household able to produce a given crop with fewer days of work could expand farm production, reduce working hours, or use some of the time elsewhere. Earning more outside farming requires another productive activity that the household can reach.

The World Bank and the Development Research Center of China's State Council describe the expansion of township and village enterprises as an important part of this transition. From 1978 to the mid-1990s, these enterprises provided nonfarm jobs for large numbers of rural workers. Employment close to home also allowed households to combine farming with wage work rather than requiring an immediate, complete departure from agriculture. [15, p. 18]

The availability of labor alone did not establish those jobs. The same historical account connects enterprise expansion to demand for consumer goods and to local governments' incentives to develop production. Organizations and markets capable of using workers' time were part of the transition. [15, p. 18]

A village workshop illustrates the connection. A household may have time for additional work while the workshop lacks finance to purchase materials. Even with materials, the workshop needs buyers. Agricultural productivity creates a possibility whose realization depends on capacities outside farming. An improvement in one activity creates a possibility that other capacities may be needed to realize. The workshop needs finance, organization, and buyers before the household's available time becomes productive nonfarm employment. Similarly, workers released from an automated task need an accessible destination for their work. How quickly that destination becomes available is a further part of the transition.

4.6 Capacity building and adjustment run on different clocks

The timing problem arises because installing equipment, learning to use it, and finding new roles for workers take different amounts of time. In an automation project, some workers might move into revised roles immediately while others need training. Someone seeking a first job may encounter a changed hiring pattern while the firm and local labor market are still adjusting.

These processes run on different schedules and involve different people. A firm may resume expansion while displaced workers are still seeking employment. Later jobs may go to new entrants rather than those who lost earlier opportunities. Following adjustment requires tracking these transitions as well as the firm's eventual output.

Resources available during that interval affect who can reach the eventual opportunity. Two workers needing a year of training could face very different choices: one has savings and access to a suitable course; the other must take immediately available work and cannot attend. What each can do next depends on the means of crossing the transition, not merely on the new job or skill available at its end. This dependence on combinations brings us back to the distributional question.

4.7 Why does the outcome distribution not mirror its components?

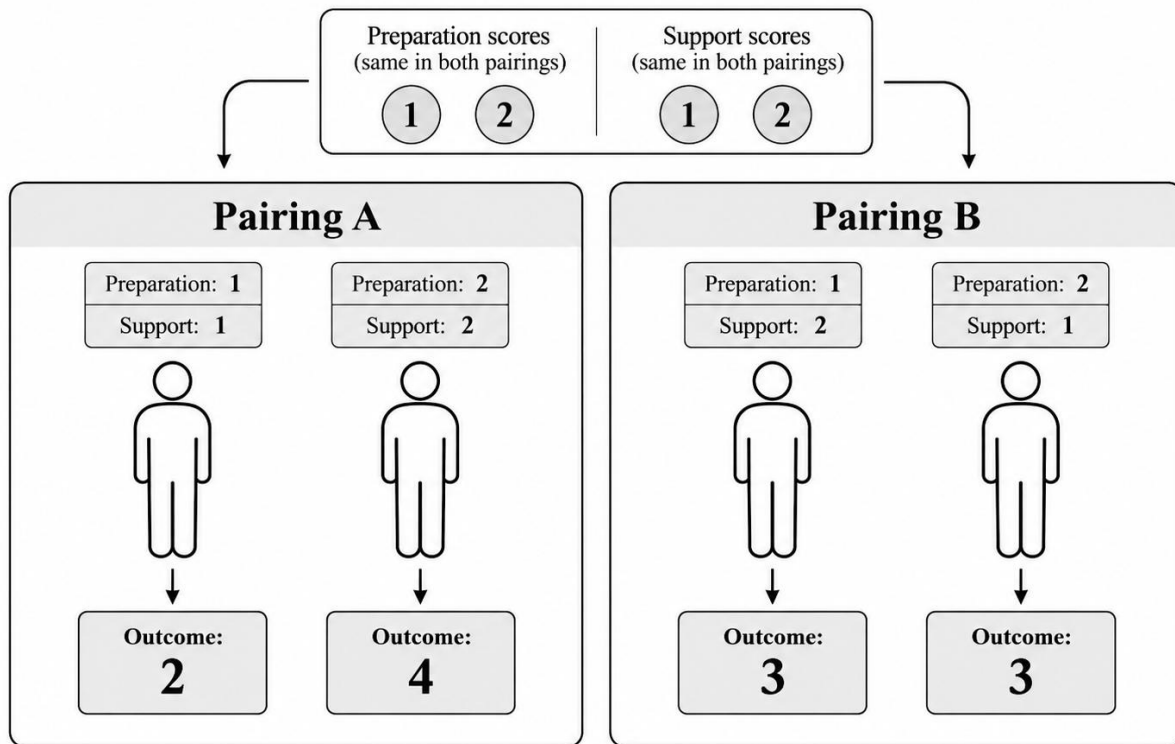
What a productive arrangement can accomplish and how its proceeds are divided jointly shape the distribution we observe. We can now follow the steps between the two business owners' characteristics and their earnings during the same year, before either business closes. First we need to establish which resources each can use, then what those resources make possible, and finally how productive activity becomes a reward to a particular person.

Knowing the separate distributions of skill and equipment leaves the first step incomplete. The more skilled owner might also have better equipment, or equipment might compensate for a difference in skill. Who is paired with what affects the combination available to each owner, even before we ask how the combination works.

To isolate that pairing effect, we give two people preparation scores of 1 and 2 and the available amounts of support scores of 1 and 2. We define an outcome score as preparation plus support. Pairing the lower values together and the higher values together gives outcomes of 2 and 4. Pairing lower preparation with higher support, and higher preparation with lower support, gives 3 and 3.

Both pairings have the same separate component distributions, the same outcome rule, and the same total outcome. **Only the pairing changes, yet the outcome distribution changes with it.**

Figure 1. Same components, different pairings



Preparation + support = outcome in both panels. The totals are unchanged; only the pairings differ.

The *joint distribution* describes how the characteristics occur together—the information missing from their separate distributions. In the business example, it concerns which owner has which combination of skills, equipment, finance, and customer access. Such access can come through relationships as well as ownership: a reliable shared facility could supply equipment to someone unable to purchase it.

Knowing the combination still leaves its productive effect to be explained. Our addition rule kept that effect simple so that only the pairing changed. In a production process, a skilled worker and a functioning machine may both be needed to complete an order: one makes the other useful. This is productive complementarity, a different relationship from pairing alone. Organization makes the combination usable by bringing the worker, machine, materials, and schedule together. Pairing concerns which inputs meet; complementarity concerns what their meeting allows them to accomplish.

Even a usable productive combination needs an opportunity to earn a return. Two businesses might submit proposals that perform almost equally well in a customer's trial. For an order of fixed value, awarding the whole order to one supplier turns a small performance difference into a sharp revenue difference. Dividing the order, where feasible on the same terms, would distribute that revenue differently. The selection rule can therefore amplify a small difference even when the ranking of the proposals is accurate.

Selection determines who receives the order; scale helps determine the amount at stake. Additional equipment and employees could let the selected firm fulfill a much larger order, increasing its revenue without a corresponding increase in the owner's personal skill. The revenue then reflects the size of a

jointly organized activity. To explain the owner's income, we must take the next step and examine its costs and the claims on what remains.

Workers might receive agreed wages while the owner receives the proceeds remaining after wages and other costs are paid. For a given contract with those payments held fixed, additional revenue goes to the owner; sharing the surplus would distribute the same proceeds differently. Expanding production would also require accounting for any additional costs. The owner's income thus depends on both the scale of the activity and a claim over its net proceeds, rather than measuring the owner's contribution alone.

These decisions also connect one firm's outcome to another's. The order awarded to one supplier is unavailable to the other; in the French robot study, adopters' expansion at competitors' expense made that connection visible. [29] The total demand available need not stay fixed as production changes. Customers might buy more when lower costs make a product more affordable, or a manufacturer might supply a previously unavailable product. A complete account therefore follows changes in the amount of activity as well as its location and division.

The distinctions identify different ways to moderate unequal outcomes. Shared equipment changes who can use a productive input; accessible training changes who can acquire a needed skill; dividing an order or its proceeds changes the allocation of returns. These actions intervene at different points between a person's starting position and the benefit received. Their effects cannot be inferred from the shared intention to reduce a gap.

An outcome distribution reflects which capacities and resources occur together, what those combinations make possible, and how access, selection, scale, and claims turn activity into benefits and burdens. Because these stages do different work, a modest capability difference can coexist with a large reward difference, while another arrangement can narrow it. Explaining an observed distribution requires evidence about how the stages operate together, including the histories that made the present combinations available.

4.8 What history lies inside the snapshot?

The equipment and customer access that enter this year's explanation were acquired somehow. Equipment might have been bought with earlier proceeds, shared through an association, or obtained before the owner entered the business. Current equipment helps explain what the business can do now; its acquisition history explains how that capacity became available. Following acquisition, use, and retained proceeds connects the present position with earlier activity and what can follow.

Following that sequence joins allocation to the development of capacity. Receiving an order can finance equipment or training; the resulting capacity can alter which later orders are attainable and what can be earned from them. Continued use also depends on support that can fail: a customer may withdraw, equipment may break, or funding may be interrupted.

Those interruptions make the histories of apparently similar positions consequential. Comparable productive capabilities can meet different events, and the same event can encounter different reserves, obligations, and options. We next follow how these encounters preserve one business's capacity while redirecting or ending another's activity.

CHAPTER 5

Why Do Similar Starting Points Diverge?

A business can retain working equipment, skilled employees, and a sound production method yet cease operating when it cannot finance the next order. The capacity described in the previous chapter depends on funding and other support that keep it usable.

Two manufacturers entering a downturn with similar equipment and recent sales may have different reserves, lenders, customers, and payment deadlines. One might continue operating while the other closes. Their responses and the decisions of others can create further differences during the downturn.

The financial crisis of 2007–2009 gives this problem a wider setting. Housing and mortgage losses developed into financial disruption far beyond the original loans. [32] Following that expansion means examining how a disturbance reaches particular firms, what responses remain available, and how their timing redirects the path that follows.

5.1 A loss does not tell us who can absorb it

Two organizations each hold assets worth 100. One owes 60, leaving its owners with equity of 40. The other owes 95, leaving equity of 5. Hold those debts fixed while both portfolios fall in value to 96. Each organization loses 4, but the first organization's equity falls from 40 to 36, while the second's falls from 5 to 1. The same asset loss removes one-tenth of the first equity cushion and four-fifths of the second.

This is the arithmetic of *leverage*: financing assets with debt magnifies the percentage change in the owners' remaining stake when asset values change. Equal assets can therefore coexist with very different cushions against loss. The ability to continue also depends on when payments fall due and whether resources can be turned into cash in time.

Even an organization whose assets exceed its debts may expect payments over several years while having a large borrowing due tomorrow. Meeting that obligation requires cash from reserves, replacement funding, or an asset sale. An otherwise comparable organization whose borrowing remains in place for several years has more time to respond. Loss-absorbing capacity and the ability to make an immediate payment are connected but distinct aspects of resilience.

A resource position therefore includes claims against the resources, contractual deadlines, and access to other organizations. Similar productive capacity can be sustained or lost through different obligations and sources of support. When that support comes through financial contracts, difficulties on another organization's balance sheet can become part of the position a firm itself encounters.

5.2 Connections can spread losses—and change their size

Mortgage finance connected balance sheets in this way. Mortgages could be pooled and their payment streams sold as securities, a process called *securitization*. Financial institutions could then use securities as collateral for borrowing. These arrangements linked the original loans to investors and funding

markets, so a problem with the loans could matter both to an asset's value and to the funding obtained against it. [33]

One important form of funding was the repurchase agreement, or *repo*: economically, a collateral-backed loan, often for a short period. Gorton and Metrick document rising collateral discounts, known as *haircuts*, in the repo transactions they study. Larger discounts meant that borrowers could obtain less funding against the same valued collateral. Their evidence connects concerns about financial trading partners' ability to meet their obligations and about collateral markets to pressures extending beyond subprime mortgage assets. [33]

The conditions of that funding differed across market segments. In *tri-party repo*, a third institution manages the collateral and carries out the exchange of cash and securities for the borrower and lender. Examining this segment gives a separate test of how broadly the funding pressure spread. [34]

Copeland, Martin, and Walker report comparatively stable collateral margins and funding for most borrowers in tri-party repo, while Lehman's funding in that segment fell sharply in September 2008. They challenge the claim of a system-wide repo run. [34] This is a scholarly disagreement about the reach of the repo-run interpretation, informed by evidence from different parts of the market. Rising haircuts and outright withdrawal of funding are distinct routes to pressure; their prevalence determines how far either account can explain the crisis. [33, 34]

Either route can leave a borrower needing cash and considering an asset sale. When many constrained holders sell into a market with limited buyers, prices can fall further. Lower prices weaken other holders' balance sheets and borrowing capacity, prompting additional sales. Brunnermeier explains this interaction between funding difficulties and asset sales as a mechanism of crisis amplification. [32] Selling may relieve one institution's immediate shortage while worsening others' access to funds: a response to the original loss becomes part of the disturbance faced elsewhere.

Connections can also help absorb losses, which makes amplification a process to investigate rather than a necessary effect of being connected. In the theoretical model developed by Acemoglu, Ozdaglar, and Tahbaz-Salehi, greater financial interconnectedness helps absorb sufficiently small shocks but transmits distress when shocks exceed a threshold. [35] The result depends on shock size under the model's conditions. In following the crisis, we must establish which connections absorbed pressure, which transmitted it, and how the responses changed the pressure itself.

5.3 Why firms and workers outside finance were affected

Funding pressure reaches a nonfinancial firm when a lender can no longer supply the credit on which production depends. A manufacturer may need to buy materials and pay workers before customers pay for completed orders. Its production method can remain sound while its financing becomes unavailable. The lending relationship is therefore part of what keeps the factory operating.

Chodorow-Reich examines this connection using banking relationships and employment data for approximately 2,000 nonfinancial firms. Firms connected before the crisis to less healthy lenders were subsequently less likely to obtain loans, paid higher interest rates when they borrowed, and reduced employment more. The employment effect was evident for small and medium-sized firms in the sample, but was not established for the largest or most transparent firms. [36]

The study uses differences in lender health to investigate whether credit availability affected employment. It also documents costs to borrowers of changing lenders, which could make an existing relationship difficult to replace promptly. [36] A relationship that ordinarily supported production could thus transmit financial distress to a firm with functioning machines and skilled workers.

A manufacturer could retain its lender and still lose business through customers' reduced spending. Mian, Rao, and Sufi find larger spending responses to housing-wealth losses in U.S. areas with poorer and more leveraged households. The distribution of debt and wealth losses helps explain the uneven decline in consumption. [37] This is a demand route to pressure on firms, distinct from the credit-supply route.

To connect that demand route to employment, Mian and Sufi examine household housing-net-worth losses across counties. Counties with larger losses experienced larger employment declines in businesses serving local demand. Their analysis examines competing explanations, including construction exposure and credit-supply tightening. Within the period studied, it finds little evidence that the hardest-hit counties quickly offset the shock through outward migration, wage adjustment, or expansion of employment producing tradable goods. [38]

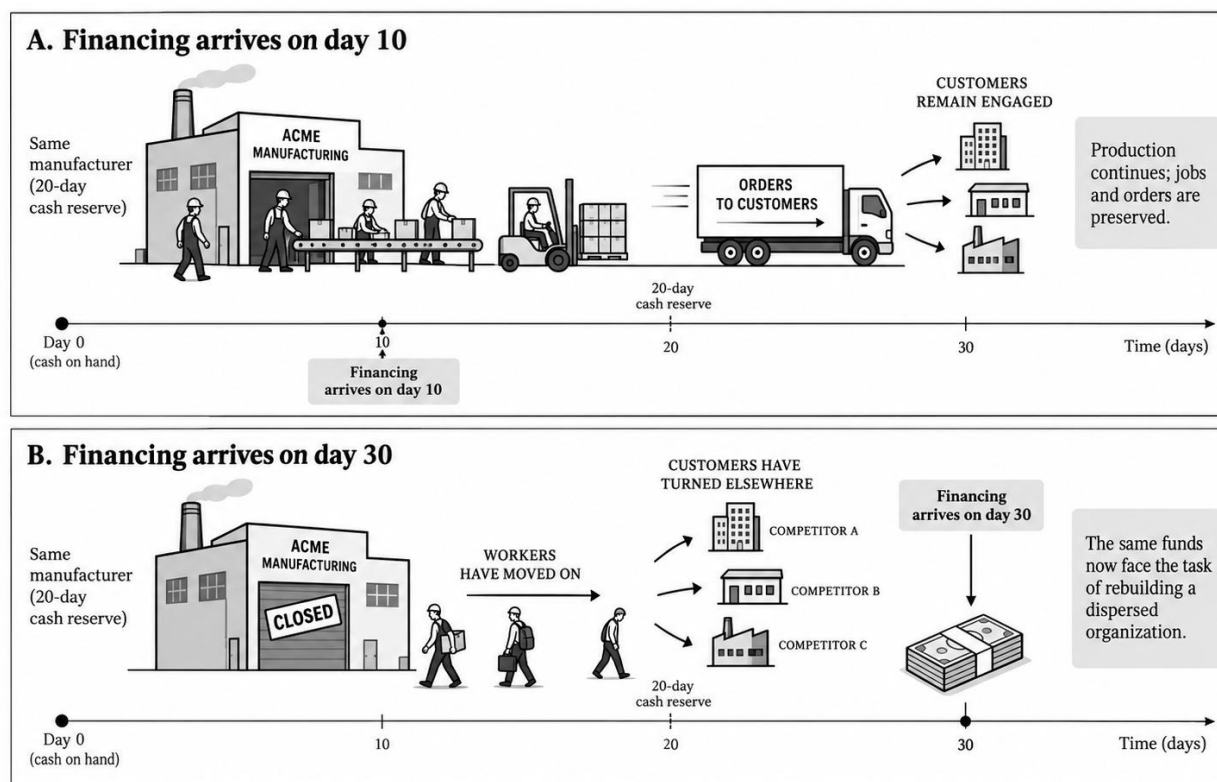
The manufacturer could face a funding problem, a loss of customers, or both. The studies investigate distinct links between financial disruption and activity outside finance; their estimates cannot simply be added into a complete account of the crisis. [36–38] For a particular firm, we need to establish which relationships transmitted the pressure and what alternatives remained.

5.4 The order of events changes what remains possible

A cash reserve can shrink while a factory still appears to operate normally; the amount remaining sets a deadline for response. With enough cash to maintain production for twenty days, financing received on day ten might preserve jobs and allow existing orders to be completed. The same financing on day thirty might arrive after closure, when workers have found other jobs and customers have made alternative arrangements.

The earlier financing supports continued operation; the later financing faces the task of rebuilding an organization whose relationships have dispersed. The amount supplied can be identical while the task changes. A ten-day financing decision also has different consequences for a business with twenty days of reserves and one with only two.

Figure 2. The same support, at different moments



In this hypothetical comparison, the same financing arrives before or after a twenty-day cash reserve is exhausted. Keeping an organization operating and rebuilding it after closure are different tasks.

Public responses can be examined through the same question: which constraint does a measure address while an activity can still be sustained? During the financial crisis, the Federal Reserve introduced facilities to provide short-term liquidity to financial institutions and support key credit markets. These interventions targeted funding and market-functioning problems. [39]

The capacity to absorb losses required a different assessment from immediate access to cash. In May 2009, U.S. banking agencies published the results of the Supervisory Capital Assessment Program for the nineteen largest bank holding companies. The assessment estimated losses, available resources, and additional capital buffers under an adverse economic scenario. Its stated purpose included sustaining lending even if the downturn became worse than expected. [40]

Access to funds, resources for absorbing losses, and information about possible future losses address different aspects of a bank's position. To assess a response, we must identify which aspect it changed and whether it did so while the relevant options remained available. Public measures and private adjustments occurred together, so improvement after an announcement alone cannot establish a program's effect. Even an effective response to one constraint leaves the recovery of other parts of the system to be followed.

5.5 One crisis does not have one ending

Relieving an immediate funding shortage is one turning point, not a restoration of every position changed by the crisis. A lender could regain market access while borrowers still lack customers, and renewed

production could precede rehiring. The different connections through which pressure spread also create different measures of recovery.

The U.S. recovery illustrates this difference in timing. The recession ended in June 2009 according to the dating reported by the Bureau of Labor Statistics, but unemployment continued rising and reached 10.0 percent in October. Payroll employment reached its low point in February 2010. [41] An end to the economy-wide contraction and an end to employment losses mark different changes in the trajectory.

Bank recovery likewise left the flow of credit to borrowers as a separate question. In May 2010, Bernanke reported improved financial positions and market access among the banks assessed in the 2009 stress exercise. In the same speech, he acknowledged that bank lending continued to contract and that the intended return to a better lending environment had not yet been achieved. Stabilizing institutions and restoring credit flows were separate accomplishments. [42]

Recovery therefore needs a specified measure and benchmark. Resumed funding, renewed hiring, and restored household savings concern different parts of the history. A business that regains its pre-downturn sales after three years reaches one benchmark. Whether it also recovers the growth it would otherwise have achieved, and whether the same workers share in renewed activity, remain further questions.

5.6 Timing matters within a life, not just within a crisis

The same calendar date can also arrive at different stages of people's lives. Chapter 1's Moving to Opportunity study found larger gains in later college attendance and earnings for children who moved when young, with estimated benefits declining as age at move increased. The researchers consider both exposure to a better environment and the disruption of moving. [5] Here the relevant clock measures the developmental experience remaining after the event, not the elapsed duration of a crisis.

Workplace adjustment has its own version of that timing problem. Training completed while someone remains in an organization could support a move into a new role; the same training offered after departure would have to be combined with a search for another employer. The resource supplied can be comparable while the surrounding relationships needed to use it have changed.

A worker entering an occupation just before its opportunities contract may reasonably describe the timing as bad luck. The contraction can arise from decisions made elsewhere while remaining beyond the worker's control. Luck here identifies an unchosen encounter with an event whose causes can still be investigated. Explaining its consequence requires tracing the event, its timing, and the resources and options available to the person affected.

5.7 What carries an earlier event into a later period?

The loss of those relationships also explains why an event can matter after it is over. Closure can leave the manufacturer with a dispersed team and broken customer connections. New financing may make production possible again, but reassembling the organization takes further work. The earlier interruption persists through the costs of rebuilding what it altered.

This directs attention to the way earlier activity changes the cost and attractiveness of later options. The literature on *path dependence* examines such changes. Pierson develops one influential account through increasing returns: taking a path can create investments, learning, and coordination that make continuing

along it more attractive and reversing it more difficult. Timing and sequence help explain which path develops. [43]

Persistent weakness can have another explanation: customers may still be spending less. Distinguishing continuing adverse conditions from a consequence carried forward by a completed shock requires evidence about what changed, what persisted, and how it affected later activity. Persistence alone leaves these explanations unresolved.

What a crisis leaves behind can include a deliberate response as well as lost resources or continuing pressure. In his 2010 account, Bernanke described incorporating lessons from the stress assessment into ongoing supervision, including more cross-firm comparisons and forward-looking analysis. The changes he described concerned supervisory methods, information, and working arrangements. [42] Their adoption changes how subsequent decisions are made; whether the new practices are effective remains a separate question.

5.8 Divergence is a process to explain

The manufacturers' equipment and initial sales describe only part of their ability to continue. A lender's condition, a customer's lost income, or the arrival of support can make the same productive resources usable in one firm and unusable in another. The path develops as these events meet particular obligations and options.

Some of those differences were present before the downturn; others arose through the responses it provoked. Closure can then create further differences by dispersing a team or consuming reserves. Comparing endpoints records all these stages together. Following the sequence helps distinguish an initial vulnerability from a pressure transmitted through others and from a consequence produced during the response.

After the immediate episode, depleted reserves, dispersed working relationships, and revised practices can shape what happens next. **An outcome can change the resources, relationships, and responses available for subsequent activity.** This connection operates during ordinary activity as well as during a crisis. We next examine how the work of one season alters the conditions for another.

CHAPTER 6

How Do Outcomes Become Later Conditions?

An imagined farming community prepares for another growing season. Some households have money for repairs; others are short of funds. A maintenance dispute remains unresolved. Farmers at the end of the canal remember whether water reached them when it was promised. The harvest is over, but its material and organizational consequences are still present.

Research on Nepal's irrigation systems helps focus our imagined community on a particular relationship. Ostrom and Gardner examine farmers near a canal's water source and those farther downstream, including how their different positions affect bargaining over maintenance and water use. [44] The physical need for a working canal joins participants whose interests in its operation can differ.

A successful harvest, an agreed repair, and a routine allocation can each change what those participants are able and willing to do next. By following one season into another, we can examine how operating the arrangement alters the resources, expectations, and relationships through which people maintain or revise it.

6.1 The next round does not begin from the same starting point

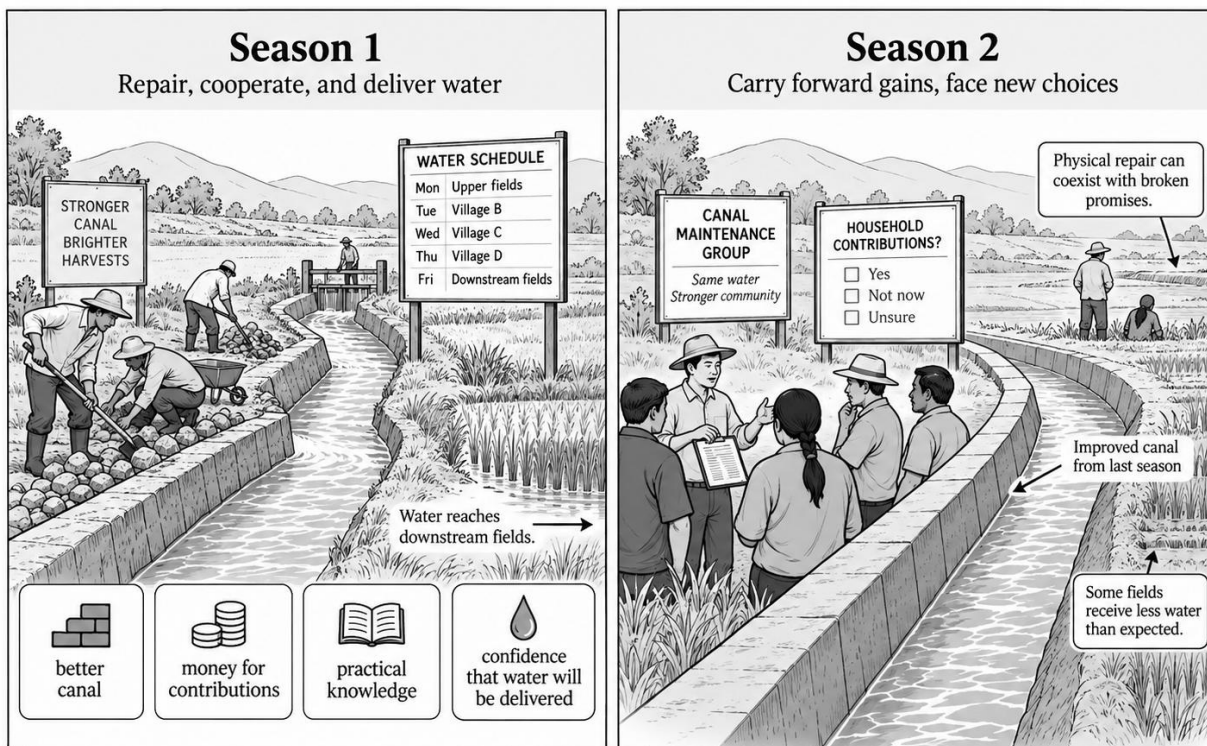
In our community, farmers agree to repair the canal and follow a water schedule. They complete the work, respect the schedule, and deliver water to downstream households during a dry period. The next season may then begin with a better canal, more money available for contributions, and greater confidence that a water claim will be honored. The organizers may also have learned how long repairs take and whom to involve.

These consequences support different parts of the next season's work. Money can finance a contribution, confidence can sustain willingness to contribute, and practical knowledge can help contributions accomplish the repair. A purchased tool also remains usable even by someone who does not know how its purchase was financed. Earlier outcomes enter later activity through material resources as well as through memory and judgment; continued provision depends on how those consequences work together.

Physical and organizational gains can also pull apart. If the repair is completed but promised water fails to reach downstream fields, the canal's physical condition may improve while some farmers become less willing to maintain it. Another household's larger harvest could increase its ability to contribute. The next collective decision would then bring together participants whose resources and expectations had developed differently.

Feedback occurs when the consequences of an activity return to affect its later operation. Here, water allocation affects future maintenance participation, which affects subsequent water delivery. **The conditions of the second season are partly produced during the first.**

Figure 3. What one season leaves for the next



In the imagined community, repairs, income, and experience carry into the next season. A stronger canal can coexist with uneven water delivery and different willingness to contribute.

6.2 What makes cooperation worth repeating?

To understand why participation might continue despite these different positions, we need to examine how the farmers organize their obligations. In comparative research on Nepal, Ostrom, Lam, and Lee reported that farmer-managed systems tended to achieve higher average performance than state-operated systems, despite often using simpler infrastructure. [45] The comparison makes organization, rather than engineering sophistication alone, a relevant part of the explanation.

Enforcement provides one way to investigate that organizational difference. Bastakoti and Shivakoti found stronger enforcement in the farmer-managed systems they examined, alongside more favorable conditions for collective action and performance. Their analysis distinguishes rules developed by users from rules largely formulated by external officials. Self-governance here involves making and applying rules for shared activity. [46]

The upstream and downstream positions help explain why rules matter. An upstream farmer might obtain water while contributing little maintenance work. A downstream farmer might contribute substantial work yet receive unreliable water. Both depend on a functioning canal, but their immediate opportunities and obligations differ.

A dependable bargain has to connect those unequal positions. In our community, a maintenance obligation linked to a recognized water claim could make participation worthwhile if both the contributions and the promised delivery are enforced. Records of work, checks on water delivery, and a way to raise disputes would let participants assess whether the bargain was honored. A household that

repeatedly works but loses its water turn has grounds for questioning the arrangement even while the written schedule remains in place.

Receiving promised water can sustain willingness to maintain the canal; repeatedly losing it can weaken that willingness. Contribution records, delivered water, and responses to complaints give farmers grounds for confidence or doubt. What they make of that experience matters for the next season.

6.3 Experience has to be interpreted—and carried forward

A better harvest after a revised water schedule leaves farmers with an interpretive problem. The improvement might reflect the schedule, more rainfall, fewer unrelated crop losses, or a combination. Farmers still have to decide whether to retain or revise the schedule while some of those causes remain uncertain. What they take the experience to mean can therefore affect next season's practice even when their inference is mistaken.

For that interpretation to survive beyond the people who formed it, it must enter a practice others can use. Levitt and March describe organizational learning as the preservation of lessons from experience in routines that guide subsequent behavior. Routines can transmit practical knowledge across changes in personnel, but they can also preserve incomplete or mistaken inferences. [47]

A lesson might become a pre-season inspection, a revised contribution record, or an established way of hearing complaints. A new organizer could then follow the practice without witnessing the original failure. Making the lesson durable requires more than remembering it: someone has to adopt the revised procedure and give it a place in the organization's work.

An episode in the Kerabari irrigation system shows dissatisfaction entering an organizational decision. Ostrom and Gardner report that farmers in one branch considered those in the other insufficiently active in maintenance and operation. Users replaced a single committee with separate branch committees while retaining a common chairperson who owned land in both areas. [44, p. 108] This documents the organizational response, rather than estimating the committee change's separate effect on performance.

The Kerabari response shows that recognizing a problem and changing an arrangement are different steps. A complaint must reach people able to hear it and act, which raises the question of who has influence.

6.4 Who can turn an outcome into influence?

The ability to secure a hearing can differ even among farmers reporting the same failure. Separate complaints about unreliable water could become a coordinated demand through representation, records of missed turns, and negotiation. Those activities could give downstream farmers a stronger voice in the committee. An upstream farmer might instead hold more land, control an important intake, or have closer relationships with committee members. These are different routes by which a position can affect whose evidence is considered and which proposals can be acted on.

We use *power* here to mean the ability to shape relevant decisions and practices. To explain its exercise, we must trace how resources, authority, relationships, or coordinated action affect a decision. Possessing a resource and using it to influence a rule are distinct steps. An earlier outcome enters this explanation when it changes the resources or organizational means available for taking the second step.

Successful cooperation could, for example, leave our farmers with an association, reliable records, and negotiating experience. They could use these results to challenge a later proposal, even when the proposal concerns more than the water delivered last season. Repeated failure could instead disperse participants and weaken their ability to hold a committee accountable. The outcome changes their capacity to press a claim, not just the claim they would like to press.

Effective influence can also persist when formal authority changes. Acemoglu and Robinson model a group that loses formal political power but invests in other means of influence and preserves favorable economic institutions. Their model shows how formal authority and effective influence can develop differently under specified conditions. [48] It provides a reason to follow the means through which decisions are shaped rather than read a formal change as a complete account of who can act.

Policies can themselves change those means. Pierson's account of policy feedback follows how policies reshape subsequent politics through resources and incentives, and through information that affects how people understand their position. [49] The relationship now runs in both directions: influence shapes a policy, while operating the policy can change who has the resources, organization, or information to influence its future.

A program that helps form user associations may therefore face different politics when its continuation is considered. The groups might support it, seek different terms, or oppose its administration. Interests become effective voices only through organization and access to decisions.

6.5 Local experience can change wider institutions

Access to authority also helps determine how far a local result travels. Chapter 2 followed the Household Responsibility System's development through experimentation, local protection, and wider acceptance. Deakin and Meng describe how local and provincial support helped the Xiaogang arrangement survive, while reports and political interpretation brought it wider attention. [14] We now follow the link from a reported outcome to people able to permit change beyond the original participants.

A report made an earlier outcome available to a new audience. Officials could use it in a policy argument; wider recognition could then alter the risks facing others considering a similar practice. In this reading of the historical account, local initiative gained wider reach through protection, interpretation, and public authority. [14]

The evidence that a reported success influenced adoption must be distinguished from evidence about what caused that success. McMillan, Whalley, and Zhu examined the contributions of price changes and changed incentives, attributing a substantial share of the measured productivity increase to payment-system changes. [12] That attribution concerns productive effects; the historical account concerns how reported experience helped secure support for reform.

The canal shows the same intermediate steps on a smaller scale: people interpret an outcome, communicate it, obtain a hearing, and act through a body able to revise a rule. Local success does not travel by itself; people and institutions carry it.

6.6 Feedback can alter the bargain that sustained it

A rule may also need revision because a successful technical change alters the bargain it supported. A more durable water-diversion structure could improve our canal while reducing upstream farmers' dependence on the downstream labor previously needed to rebuild it. That reduced dependence could weaken their incentive to honor the agreement through which they obtained the contributions. The improvement changes the relationship within which the existing rule operates.

Lam's analysis of technological investment and irrigation governance in Nepal develops this concern: technical changes can intensify asymmetries among users and weaken incentives to contribute to maintenance. Engineering improvements alter the dependencies within which existing rules operate. [50]

The appropriate response may therefore concern both the structure and the bargain around it. Revising water obligations and maintenance contributions together could address the altered dependence. Simply preserving the formal rule could leave participants with a different practical relationship from the one that made compliance worthwhile before the improvement.

Whether they can revise the bargain depends on the organizational means examined above. Downstream farmers might withdraw from maintenance after repeatedly missing their turns; worse delivery could encourage further withdrawal and reinforce decline. Monitoring, negotiation, or replacement of an ineffective organizer could instead interrupt the sequence. Resources, interpretations, relationships, and authority help determine whether dissatisfaction becomes withdrawal or an effective attempt at repair.

A technical improvement can therefore change the bargain on which its continued operation depends. Maintaining the improvement may require revising obligations as well as repairing physical infrastructure.

6.7 How do we know that an outcome changed what followed?

The mechanisms just described tell us what to look for, but continued cooperation is not sufficient evidence that any one of them operated. A community might cooperate in two seasons because unchanged rules support both, because water remains abundant, or because households more able to cooperate stay while others leave. To attribute the continuation to an earlier outcome, we must identify what that outcome changed between the seasons.

An explanation through changed expectations requires evidence about what farmers believed before and after the first season, how those beliefs affected their decisions, and whether contributions changed. Interviews about confidence and records of attendance or repairs answer different parts of that inquiry.

A material explanation requires a different observation. If a larger harvest enabled later maintenance payments, the link runs through household funds and subsequent contributions, whether or not confidence changed. A rule-revision explanation follows the complaint, the hearing, the decision, and the practice adopted. These accounts can work together, but each needs evidence about its own intermediate steps.

Longitudinal research can place those steps within a continuing history. Lam and Ostrom studied irrigation assistance initiated in Nepal in the mid-1980s using observations from three periods. Their analysis followed combinations of infrastructure, organization, rules, sanctions, and collective action, concluding that infrastructure investment alone was insufficient to sustain higher performance. [51] The

study connects an intervention with later performance across changing conditions; it is not a randomized estimate of one feedback mechanism.

To identify feedback, we must show what an earlier outcome changed and how that change affected later activity. Continued operation alone cannot distinguish adaptation from unchanged support—or from an arrangement participants cannot alter.

6.8 How do advantages come to occur together?

Following these intermediate steps also helps explain why advantages come to occur together. A successful season can leave the same household with funds for repairs, confidence in cooperation, and a stronger role in the organization. Another household can lose resources and confidence together. The later combination is partly a joint product of earlier activity, rather than a collection of characteristics whose association must simply be taken as given.

The first business's contract illustrates how one result can build several parts of a position at once. Its proceeds can finance equipment, the work can develop experience, and the completed order can supply a reference. Those resources can then work together when the owner seeks the next contract. The second owner might improve a skill while drawing down savings, or regain access through another customer or shared equipment. The history explains both why some supports reinforce one another and why improvement in one component may coexist with deterioration in another.

Continued participation supplies another link in that history. In the Dutch research-funding study, an initial award affected whether closely ranked applicants entered later competitions. [2] An allocation can therefore influence later results by changing who pursues another opportunity, as well as by financing resources used within it. Participation belongs in the account of how later combinations form.

The resulting combination can also pass to someone who did not produce it. An income-producing asset retained by a business owner and transferred to a child changes the child's resources without first requiring the child to repeat the parent's activity. Rights of retention and transfer give a prior outcome that reach. A comparison of present skills and assets would record the transferred advantage; its explanation requires following the transfer.

Resources gained through cooperation can also support demands for different rules, changing who can use a resource or claim its proceeds. Earlier activity can therefore alter both the positions entering a later comparison and the arrangements that make those positions consequential. The irrigation story brings these routes together; the next illustration isolates just one of them.

6.9 How can ordinary variation produce a skewed distribution?

A different question is how unequal holdings can arise even when participants begin alike and follow the same rules. Imagine many people each holding 100 units of an asset. In every round, each experiences a proportional gain or loss, and the result becomes the holding for the next round. A 10 percent gain adds 10 units to a holding of 100 but 100 units to a holding of 1,000. Earlier differences therefore change the size of later gains and losses.

Repeated proportional changes multiply rather than add. We choose the same probability distribution for everyone, with independent draws across people and rounds. Different sequences nevertheless produce

different holdings. The calculation note describes the normal log-growth influences used to make this distribution exactly calculable.

The process is designed so that each participant's expected holding remains 100 even as the distribution widens. This expectation is over possible sequences; a finite group's realized total can vary. The centering adjustment and full derivation appear in the calculation note. Table 1 gives the resulting population percentiles.

We choose the multipliers so that each participant's expected holding remains 100. This lets us examine widening dispersion without also increasing the distribution's mean. The expectation concerns the range of possible sequences; the realized total in a finite group can vary.

Table 1. Holdings generated by an illustrative accumulation process

Completed rounds	Median holding	99th-percentile holding	99th percentile / median
0	100.0	100.0	1.00
1	98.0	156.1	1.59
10	81.9	356.6	4.35
50	36.8	987.4	26.84

Calculated population percentiles under the stated assumptions, not empirical observations or simulated sample results. Holdings are in hypothetical units; a round is not a specified calendar period. The 99th percentile is the level below which 99 percent of holdings fall, not the average holding of the top 1 percent.

After fifty rounds, the 99th-percentile holding is 26.84 times the median. Under the stipulated mean-preserving process, the median falls while relatively rare large outcomes keep the mean at 100. The example shows widening dispersion without any difference in initial ability or in the growth rule. It is a mathematical illustration, not an estimate of a real population.

At each positive number of completed rounds, the logarithms of holdings are normally distributed, so holdings follow a *lognormal distribution*, which is right-skewed with a long upper tail. [52] The distribution of each round's influences is unchanged; their repeated operation on accumulated positions widens the resulting distribution. These are finite-time lognormal distributions, not a Pareto power-law tail. Research on wealth distributions examines other processes that can generate skewness and thick upper tails. [1]

Widening depends on participants experiencing different proportional changes and retaining each result for the next round. Identical proportional changes for everyone would preserve their holding ratios. Different sequences can also let someone ahead fall behind, so greater dispersion does not determine who remains at the top. The distribution's spread and movement among positions are different features of the same process.

Repeated proportional changes act on holdings carried forward from earlier rounds. This accumulation—not a difference in participants' stipulated ability—is what the illustration isolates. The next question is what happens when the retention rule changes.

6.10 What changes when the process changes?

After each round, we let participants place one-fifth of their holdings in a common pool and divide it equally, with no loss in making the transfer. Total holdings at that step remain unchanged, while every gap between two participants becomes four-fifths of its previous size. The next round's proportional changes then act on these altered holdings.

The result concerns the pooling step, whose contraction can be derived directly. Subsequent draws can widen holdings again before the next pooling step contracts the gaps then present. We have established this repeated stepwise contraction, not the modified process's long-run distribution. That distinction keeps the comparison focused on the particular operation being changed.

The same specificity matters when we return to the earlier examples. Shared equipment changes access to productive activity; a rule for dividing proceeds changes what participants retain; support during an interruption can preserve another chance to participate. Each acts on a different connection between present activity and later possibilities. Their effects depend on which constraint or claim is changed, not simply on their all being described as intervention.

The baseline example itself permits individual retention of gains. Pooling changes that retention rule while leaving the growth influences unchanged. We are comparing two specified arrangements, rather than an institutional arrangement with an absence of institutions. Their feasibility and value remain questions beyond the arithmetic of the pooling step.

Nor does a narrower observed distribution identify pooling as its cause. Even in the baseline illustration, less variable growth influences or fewer elapsed rounds would produce a lower 99th-percentile-to-median ratio. Inferring the effect of an intervention requires evidence about the transitions and the relevant alternative arrangement. The shape of the distribution is a result to explain, not a direct measure of the intervention that produced it.

6.11 What does the next snapshot record?

A later snapshot records positions produced through many kinds of transition. Holdings may change under a fixed rule; elsewhere, cooperation may change both resources and the rules governing them. The observed combination cannot tell us its history by itself.

The point-in-time view explains how current combinations and arrangements produce outcomes; the dynamic view explains how those combinations and arrangements came to be. The accumulation illustration isolates one transition. Explaining an actual distribution also requires evidence about access, selection, scale, claims, and developing capabilities.

Even a sequence of distributions leaves part of that history unspecified. Two populations could have the same four income levels every year, with the same people occupying each level in one population and people rotating through them in the other. The annual distributions would be identical, but persistence and mobility would differ. Following people, entry, exit, and relationships reveals whether the observed pattern reproduces the same positions or is compatible with movement among them.

A history detailed enough to reveal those connections also identifies possibilities for change. The canal's next season might begin with more reliable infrastructure and stronger cooperation while access remains

unequal or maintenance falls heavily on a few households. Knowing how those results are sustained helps us ask what a proposed change would alter. Choosing whether to preserve the bargain, redistribute a burden, or develop a different arrangement requires reasons about the people and possibilities affected. That is the task of Chapter 7.

Calculation note for Table 1

We specify the process used in Table 1 so that its assumptions, centering adjustment, and distributional results can be followed separately. Let $W_{i,t}$ denote participant i 's asset holding after t completed rounds. Each participant begins with $W_{i,0} = 100$ hypothetical asset units. A round has no assigned calendar duration. Ability and the growth rule are the same for everyone; consumption, debt, and entry or exit are outside the calculation.

For each participant and round, let $\varepsilon_{ik,t} \sim N(0,1)$, independently across i , k , and t . The index $k = 1, 2, 3, 4$ identifies four influences within a round; i identifies the participant, and $t = 0, 1, \dots$ identifies the starting round. We combine the influences into the log-growth value $Z_{i,t}$ and update the holding by its positive multiplier:

$$Z_{i,t} = 0.1 \sum_{k=1}^4 \varepsilon_{ik,t} - 0.02, \quad W_{i,t+1} = W_{i,t} \exp(Z_{i,t}).$$

Four is a convenient decomposition of the chosen round variance, not a count of empirical causes. The independent terms each contribute $0.1^2 = 0.01$ to variance, so their sum has variance 0.04 and standard deviation 0.2. An aggregate standard-normal draw multiplied by 0.2 would give the same growth distribution. Keeping the four terms makes explicit how several normal influences can enter the illustration without assigning them to talent, institutions, or other measured characteristics.

We subtract 0.02 to keep the expected multiplier at one, while allowing log growth to have a negative mean. For a normal variable with mean μ and variance σ^2 , the expected exponential is $\exp(\mu + \sigma^2/2)$. [52] Here $Z_{i,t}$ has mean -0.02 and variance 0.04, giving

$$\mathbb{E}[\exp(Z_{i,t})] = \exp(-0.02 + 0.04/2) = 1.$$

Without that subtraction, the expected multiplier would exceed one. Because each new draw is independent of the holding already accumulated, $\mathbb{E}[W_{i,t+1} | W_{i,t}] = W_{i,t}$. Iterating from the common starting holding gives $\mathbb{E}[W_{i,T}] = 100$. This is an expectation over possible sequences, not a rule requiring a finite group's realized total to stay fixed.

After T completed rounds, the logarithm of the ratio $W_{i,T}/W_{i,0}$ equals the sum of the T log-growth values. For $T > 0$, it is normal with mean $-0.02T$ and variance $0.04T$. Thus the holding at population quantile p , for $0 < p < 1$, is

$$Q_T(p) = 100 \exp[-0.02T + 0.2\sqrt{T} \Phi^{-1}(p)].$$

Here Φ is the standard-normal cumulative distribution function and $\Phi^{-1}(p)$ its quantile: the value below which a fraction p of standard-normal draws falls. The formula applies the lognormal quantile relation in [52]. Since $\Phi^{-1}(0.50) = 0$, the median is $100 \exp(-0.02T)$, while the expected holding remains 100. The centering removes growth in the mean, not dispersion; the widening upper portion carries enough weight to keep the mean above the falling median.

Table 1 uses $T = 0, 1, 10, 50$ and $p = 0.50, 0.99$, with $\Phi^{-1}(0.99) \approx 2.326347874$. At $T = 0$, every holding is 100. At $T = 50$, the median is approximately 36.8, the 99th percentile is approximately 987.4, and their ratio is 26.84. These are calculated finite-time distributional values, rounded for display, not simulated sample results or a stationary distribution. The 99th percentile is a cutoff, not the mean of the top one percent.

For the modification in §6.10, consider one pooling step with n participants. We suppress the time index only for this step: W_i is the holding immediately before pooling, $\bar{W} = n^{-1} \sum_{i=1}^n W_i$ is the actual group mean then, and W_i' is the holding immediately afterward. The prime denotes the post-pooling holding, not a derivative. Costless pooling of one-fifth and equal redistribution give

$$W_i' = 0.8W_i + 0.2\bar{W}, \quad W_i' - W_j' = 0.8(W_i - W_j).$$

The common redistributed amount cancels from each pairwise difference, which explains the four-fifths contraction. Summing the new holdings gives $\sum_i W_i' = \sum_i W_i$, so the pooling step preserves the realized total, unlike the random growth step. Subsequent growth acts on these new holdings. This establishes the stated contraction at each pooling step without determining the modified process's long-run distribution.

CHAPTER 7

What Counts as Improvement?

An irrigation system can sustain production while distributing water, maintenance work, and influence unevenly. Following how those results enter the next season identifies possible changes: a different contribution rule, a protected way to complain, or a repair that changes the need for labor. The explanation tells us where a proposal would act and what it might alter. Choosing among those proposals requires reasons about which gains to preserve, which burdens to reduce, and whose opportunities deserve attention.

Such reasons are needed because the relevant consequences can move in different directions. A society could become more productive, healthier, less equal, more politically free, and less able to sustain its use of natural resources over the same decade. Judging that trajectory means examining what improved, for whom, at whose expense, and with what future consequences. We can recognize the relief of a serious deprivation without treating that achievement as a verdict on every other change.

We examine distinct evaluative perspectives because they identify different reasons for recognizing a gain, questioning a burden, or preferring another course of action. These reasons can support different judgments about the same change. The connections developed in earlier chapters specify what is being judged; the perspectives considered here help explain why one change might be preferable to another.

7.1 What improved—and compared with what?

In the Oregon experiment, Medicaid coverage increased service use, reduced financial strain, and reduced positive screenings for depression over approximately two years. The study did not find statistically significant improvements in the selected physical-health measures. [18]

Financial protection and the measured mental-health gain give specific reasons to value coverage. Reduced exposure to medical bills and relief from depression matter to people's lives even when other health measures show no detected improvement. A verdict on the whole program must also consider its other consequences and costs. The first task is therefore to identify the gain without making it stand for the entire choice.

That identification also needs a comparison. A community's health could deteriorate during an economic disruption while an intervention prevents an even larger decline. Health would then be worse than before but better than it would have been without the intervention. A comparison between periods describes the trajectory; a comparison with what would have happened without the intervention assesses the intervention's contribution.

Another comparison concerns feasible alternatives. An intervention can help while another use of the resources would help more. In a local budget, financing a clinic might require postponing a water repair. The benefits of the clinic and the consequences of delayed repairs both enter the choice.

An identified gain supplies one reason for a favorable judgment; the comparison with other consequences and feasible alternatives determines how far that reason takes us. The explanatory studies contribute

evidence to this choice. Where benefits and costs fall on different people, the distribution of those consequences becomes part of the judgment as well.

7.2 Who gains—and is compensation real?

Chapter 4's French manufacturing study located gains within robot-adopting firms alongside employment losses elsewhere in the market. [29] Evaluating the change requires including the people losing opportunities as well as those receiving gains.

Take two households with annual incomes of 20 and 80, holding household size and prices fixed and considering income alone. After a change, their incomes become 15 and 105. Combined income rises from 100 to 120, while the poorer household loses one-quarter of its income. Higher combined income coexists with a loss whose significance must be assessed in that household's life.

The winner's gain is large enough to compensate the loser in this example. A costless transfer of 6 would leave incomes of 21 and 99, above both starting incomes. Kaldor proposed judging a change by whether those gaining could compensate those losing while retaining a gain. This separates a test of potential aggregate benefit from a judgment about the justice of the resulting distribution. [53]

Potential compensation and delivered compensation leave the poorer household in different positions. Before the transfer it receives 15; after the transfer it receives 21. The possibility of payment leaves the first position in place until someone makes the payment. As with a formal entitlement, identifying resources that could provide a benefit leaves the delivery arrangement to be established. An assessment invoking compensation must follow that additional step.

Other losses raise further questions about adequacy. A lost right, an unwanted relocation, or an interrupted education requires an assessment of whether compensation is possible and what could make it sufficient. Money may not make the affected person whole.

The practical choice therefore includes how an expansion is organized, whether avoidable losses can be prevented, and whether promised compensation reaches those bearing the costs. Even after a payment is delivered, its adequacy depends on what it allows the affected person to recover or do. Distribution and the use of what is received both enter the evaluation.

7.3 What can people actually do?

To assess that use, we need to look beyond the amount transferred to the lives it makes possible. Sen's question, "Equality of What?", directs attention to what a person is able to do and to how differences among people and their circumstances affect the use of resources. Equal quantities of resources can offer different practical opportunities. [54]

The treatment example makes the difference visible. Two people hold the same entitlement at the same clinic. One can travel by bus and arrange time away from work; the other requires accessible transport that is unavailable. The same coverage leaves them with unequal opportunities to receive treatment.

The capability approach uses *capability* to mean a real opportunity to be or do something, rather than a personal skill alone. An achieved state or activity is called a *functioning*. Robeyns explains why an evaluation can concern both what people achieve and the meaningful alternatives available to them. [55]

Declining a training course illustrates the distinction. One person may prefer another available activity. Another may want to enroll but be unable to arrange childcare. The same enrollment record can reflect a choice among usable alternatives or the absence of an alternative the person wants.

The treatment and training examples identify a dependency that an equal allocation can leave hidden. Needs, abilities, commitments, and circumstances determine whether a benefit is usable; what supports one person's participation may leave another person's obstacle in place. Individuality matters to the conversion from resources or entitlements into opportunities.

This gives the capacity distinction from Chapter 4 an evaluative role. A health service's capacity concerns its ability to provide treatment; a person's capability, in the sense just defined, concerns the opportunity to obtain it. The service may become more productive while particular patients still cannot reach it. Removing a transport barrier could expand a patient's opportunity while leaving clinical capacity unchanged. To judge an improvement, we need to specify whether it expands provision, usable access, or both.

Usability alone does not settle which opportunities deserve support. A greater ability to intimidate others is an expanded ability with objectionable consequences. The capability approach requires attention to the value of the opportunities being assessed. [55] An account of improvement should therefore ask both whether people can use the possibilities being expanded and why those possibilities are worth expanding.

7.4 Do narrower gaps mean better lives?

The same attention to people's lives matters when the proposed improvement is a narrower gap. The gap between our two households could narrow solely because the better-off household loses income, with no benefit to anyone else. Equality of income would increase while the poorer household's income remained unchanged. Whether the narrower gap counts as an improvement depends on the reasons for valuing it.

Parfit distinguishes concern with equality from giving priority to people because they are worse off. The latter supports helping someone in a bad position without valuing a reduction in someone else's advantage for its own sake. People can care about equality and other values together; the distinction identifies different reasons for judgment. [56]

The distinction helps us ask what a smaller gap would accomplish in the household example. Would it reduce deprivation, expand an opportunity, or alter an objectionable relationship? The answer identifies a reason to favor the change, rather than treating reduced distance between two numbers as a sufficient account of better lives.

A defense of unequal rewards through effort addresses a different question: what contribution gives someone a claim to a larger reward? Recall the workers whose access to retraining differed in Chapter 4. Completing the course required effort; being unable to afford time away from earnings constrained the other worker's opportunity to supply it. Their eventual earnings record the consequences of both effort and access.

Even an estimate isolating effort's contribution would leave the appropriate reward to be judged. Roemer's examination of equality of opportunity identifies the need for additional ethical judgments about rewards to effort. Explaining the contribution and deciding how much inequality it justifies are separate tasks. [57]

Luck also separates an explanatory contribution from an evaluative conclusion. An unchosen event can help explain a gain or loss even when its causes lie in other people's decisions. Deciding whether to insure against it, compensate its consequences, or leave the outcome in place requires reasons about responsibility, solidarity, rights, and acceptable risk. **Unequal outcomes do not, by themselves, establish unequal effort, unequal responsibility, or a justified scale of reward.** The history clarifies what happened; those additional reasons determine what we make of it.

7.5 How was the result obtained?

The size and distribution of rewards still leave open how they were obtained. Our imagined canal could deliver the same quantity of water under two arrangements. In one, maintenance duties are negotiated, contributions are recorded, and farmers can challenge a missed turn. In the other, an organizer compels work through intimidation and excludes particular households from raising complaints. Equal output coexists with a substantial difference in freedom, treatment, and participation.

Rights can constrain the means used to obtain a gain, even where violating them would increase output. Rawls gives an influential statement of this position by assigning basic liberties priority over economic and social gains. [58]

The relationships created by those rules provide another reason to examine the means. Anderson's account of democratic equality emphasizes relationships among people who can participate as equals. Subordination can make an arrangement objectionable alongside concerns about the goods people receive. [59] A protected right to complain at the canal can therefore matter both as a way to improve water delivery and as protection against dependence on an organizer's favor.

For that protection to be usable, a farmer must be able to speak without an overriding fear of retaliation. An invitation to a meeting does not supply effective voice under those conditions. Nor does majority authorization by itself settle the justice of the burdens imposed. We need to examine both how a decision is authorized and what the authorized decision permits people to demand of others.

We can likewise recognize the agricultural productivity gain attributed to changed household incentives in the China case while asking separately about security of claims and the ability to challenge decisions. [12] Those further questions require their own evidence; the productivity study does not establish harms on those dimensions. The evaluation concerns what a person receives, what the person can decide, what others may demand, and what recourse exists when a claim is denied.

This makes the institutional relationships examined earlier subjects of evaluation in their own right. They affect production and allocation, but they also establish forms of control, dependence, and participation among people. Both their effects and those relationships can be reasons for preferring one arrangement to another.

7.6 Whose future enters the judgment?

The period over which we follow those effects changes what enters the judgment. Chapter 5 showed that the U.S. recession ended before payroll employment stopped falling. [41] Ending an assessment at the first turning point would leave later employment losses outside the picture. An endpoint suitable for one measure can be too early for another part of people's experience.

A household regaining its former income after several years has reached one recovery benchmark. The intervening losses remain part of its history, along with any effects on savings, health, or opportunity. Evaluating the path requires keeping those costs visible alongside the later position. A later benefit may outweigh an earlier cost under a defensible judgment, but that judgment must include both.

The same requirement extends beyond losses already experienced to the conditions being left for future activity. Our community could raise this year's harvest by postponing essential maintenance or drawing down a water reserve that cannot be replenished. Part of the current gain would come from weakening the conditions of later production. A favorable current result would then carry a future burden.

Dasgupta's account of national wealth evaluates the productive assets, including natural assets, that support well-being over time. Fleurbaey and Blanchet likewise distinguish assessing current welfare from assessing whether it can be sustained. [60, 61] For the canal, this means examining the infrastructure and water supply left for subsequent users alongside the value of the present crop.

A forecast can clarify the likely consequences of postponing repairs. Judging whether the current farmers' additional income justifies risks to future farmers also requires reasons about need, responsibility, available alternatives, and obligations across time. Later users' interests must enter that judgment explicitly.

Because those future consequences are uncertain, we also need to distinguish the quality of a decision from the outcome eventually realized. A reasonable investment can fail, while a poorly justified gamble can succeed. Evaluation can examine the information available at the time, the understanding of risks, the voice of those bearing them, and less harmful alternatives. A fortunate result is not a substitute for these reasons, just as an unfortunate result need not show that the decision lacked them.

7.7 Do we need one score?

These judgments bring income, opportunity, rights, distribution, and future interests into the same decision. A summary measure can make patterns visible, while a decision model can state priorities and show how a conclusion changes when those priorities change. Combining the concerns numerically requires specifying what the numbers represent and how competing considerations are weighted; it does not remove the need to justify those choices.

Fleurbaey and Blanchet's discussion of alternatives to gross domestic product (GDP) develops this issue for welfare and sustainability: selecting measures requires substantive choices about what they should represent. Even a larger collection of indicators leaves their relative importance to be judged. [61]

Two irrigation proposals illustrate the task. One produces a larger total harvest. The other produces a smaller total increase but substantially improves water reliability for households with the least secure access. A combined score must specify the weight given to reliability relative to output and whether the beneficiaries' initial positions affect that weight. Those choices need reasons that readers and affected people can examine.

Some comparisons allow a clear preference without settling every priority: one option can improve an important condition without worsening another that matters. Other comparisons depend on resolving conflicts among gains and burdens. A judgment remains open to criticism when it ignores a serious loss,

applies inconsistent standards, excludes affected people without justification, or rests on a false account of the consequences.

When a comparison remains contested, it helps to identify the source of the disagreement. A dispute about what a proposal will do calls for better evidence. Agreement on its likely consequences can still leave disagreement about which deserve priority. Further measurement may expose mistaken premises, while the priority itself requires evaluative reasons.

We can still make a practical choice by eliminating clearly inferior options, protecting essential rights, and identifying a defensible preference among feasible alternatives. Revising a proposal may reduce the conflict that made the original comparison difficult. This work requires reasons sufficient for the decision at hand, rather than a complete ranking of every possible social arrangement.

The decision at hand also includes preserving the present arrangement. Postponement could leave unreliable water access and unfair maintenance duties in place. Comparing a proposal with inaction means following the consequences of both, including the burdens borne by people who continue under the current rules.

7.8 What can we responsibly call improvement?

Return to the canal with a repair and rule change that gives downstream households more reliable water, preserves adequate access upstream, reduces total repair work, and provides an effective way to challenge allocation decisions. Assume its costs are affordable, maintenance is funded, and the evidence supports the durability of these gains.

On those criteria, there is a strong case for improvement. Valuable opportunities expand, a serious disadvantage is reduced, cooperation requires less work, and future provision is sustained. The reasons are connected to the changes the repair and rule revision actually make. Other problems can remain open to criticism without negating those achievements.

A larger harvest obtained by silencing one group and imposing an unreasonable unpaid burden on another would call for a different overall judgment. The gain in output would remain, alongside losses that the increased total could not make disappear.

The point-in-time and dynamic accounts specify what those reasons concern. The former identifies current positions, claims, and usable opportunities; the latter reveals burdens along the path and conditions left for later participants. Together, they keep a current gain from obscuring how it was produced, who bore its costs, or what it makes possible next.

A defensible evaluation combines that explanation with reasons for valuing gains, accepting or rejecting burdens, and recognizing responsibilities beyond the present. Several judgments can meet those demands without becoming identical. We can recognize a gain, reject an unjust means, question a neglected loss, and seek a better alternative, while keeping the lives, relationships, and possibilities changed by the trajectory in view.

Conclusion

Return to the two businesses. The first contract could give one owner income, equipment, experience, and a customer reference, while the other owner used savings to keep searching. Their subsequent decisions matter, but so do the arrangements through which effort reaches a customer, becomes a reward, and supports another opportunity.

Why can modest differences in individual capabilities coexist with much larger differences in rewards, resources, and influence? Because performance enters a process of access, productive combination, selection, scale, and claims; the results of that process then reshape the conditions of the next round. The present position and the history that produced it are both necessary to the explanation.

How present conditions become outcomes

Skills alone do not tell us which resources people can use together. Equipment, finance, time, organization, and access can reinforce a skill or compensate for a disadvantage. A productive combination must then encounter demand, and the terms of selection determine who receives an order.

The order's scale and the claims on its proceeds shape how much reaches each person. An exclusive contract can reward a small difference in performance with a large difference in income. Sharing equipment, changing access, or dividing proceeds would act at different points in this process; none is equivalent to changing the individual's skill.

The owner's effort may be essential to fulfilling an order. But the reward also depends on the order's value, other people's contributions, ownership, and the terms on which the opportunity was awarded. Recognizing the effort and explaining the reward are connected, but they are not the same task.

How outcomes change later conditions

Present capabilities have histories. Childhood environments can influence their development, and earlier rewards can finance later equipment or training. The business's first contract is not just an item in last year's accounts: its proceeds and reference can change which contracts become attainable now.

This process need not benefit only one participant or move in one direction. Losing a contract can consume another owner's reserves and narrow the time available to seek work. Support, cooperation, or a new customer can interrupt that decline. The later gap records the evolution of both businesses, not merely the success of one.

Retention and transfer extend the history further. An asset can provide income, a reserve, or a starting resource for someone who did not produce it. Property rights determine which of these possibilities is available and how long its effects can last.

Timing and luck matter through the position in which an event is encountered. A delay that one firm can survive may close another whose cash is exhausted. Financing before closure preserves an operating team; the same financing afterward faces the work of rebuilding one. Events outside a person's control do not have consequences independent of accumulated resources and obligations.

Activity may even alter the rules of later activity. Cooperation can produce an association able to press a claim; a technical improvement can change the dependence that once sustained an agreement. The path includes reinforcement, recovery, erosion, and adaptation—not accumulation alone.

Reading a distribution through its history

The accumulation illustration demonstrates one route from equal initial holdings and identical chances to wide differences. It does not identify the mechanism behind any observed wealth distribution. Similar snapshots can conceal very different histories, including persistent rankings in one population and substantial movement in another.

Nor does a narrower distribution reveal what caused it. It may reflect different shocks, elapsed time, or arrangements governing retention and sharing. Explaining change requires following the transitions and comparing actual alternatives.

Table 2 provides a guide back to the manuscript's principal connections and examples.

Table 2. How differences become outcomes—and outcomes become later conditions

Connection	What it changes	Where the discussion develops it
Formation through experience	The abilities, expectations, and opportunities people bring into later activity.	Childhood neighborhoods and access to a move: Chapter 1, §§ 1.2–1.6.
Pairing, access, and organization	Which skills and resources become usable together, and who can realize a claim or productive possibility.	Medicaid: §§ 3.3–3.6. Preparation/support pairing and productive combinations: § 4.7.
Selection, scale, and claims	The size and division of returns, including rewards from activity involving many people.	Household production rights: §§ 2.1–2.3. Firms and contracts: §§ 4.3, 4.7.
Retention, accumulation, and transfer	Earlier proceeds become assets, reserves, or another person's starting resources; losses reduce what remains available.	Property: § 2.6. Business, asset transfer, and accumulation examples: §§ 6.8–6.10.
Timing, exposure, and response	Whether a disturbance can be absorbed, and whether assistance preserves an operating activity or arrives after it has broken up.	Financial crisis and the manufacturer: §§ 5.1–5.6.
Learning, cooperation, and maintenance	Experience affects expectations, routines, and participation that sustain or weaken provision.	Nepal irrigation and organizational learning: §§ 6.1–6.3, 6.6–6.7.
Influence and rule revision	Resources, organization, and reports of results can affect who can be heard and which rules govern later activity.	User associations and Chinese reform: §§ 6.4–6.5.

The connections also depend on one another. Access to equipment may be ineffective without finance to continue working; higher output may be hard to sustain if those maintaining the system cannot claim its benefits. A rule that secures participation can therefore help sustain the capacity from which future benefits arise.

Which possibilities should change?

An explanation is useful when it shows where a practical change could act. An agency might remove a barrier to enrollment, a lender might help a firm before its team disperses, or farmers might revise water rights alongside maintenance duties. The relevant action depends on the obstacle and on the arrangements already in place.

Changes must also remain workable over time. The irrigation research found that investment in infrastructure alone did not sustain higher performance; organization and collective action mattered. [51] Who receives a benefit and who maintains the capacity to provide it are connected questions.

No participant controls the entire process. Owners face uncertain demand, farmers cannot choose rainfall, and agencies may remove one barrier while others remain. Responsibility depends on what a person or institution can know and change, as well as on whose interests the decision affects.

Evaluation requires more than an increase in output or a reduction in inequality. We must ask whose opportunities become usable, how gains and burdens are shared, whether people have rights and effective voice, and what resources remain for the future. A favorable total does not erase an unacceptable loss along the way.

The successful owner's work is still an achievement to recognize. Understanding what made that work effective helps explain the achievement; deciding the appropriate reward requires reasons about what deserves recognition and how the proceeds of joint activity should be shared. The observed reward is part of the inquiry, not its answer.

The question reaches familiar experiences: whether effort has an opportunity, whether a claim becomes a usable benefit, whether a setback can be survived, and whether an achievement can be sustained. Each calls for an account of the relationships and history behind an observed result.

How we get where we are depends on individuality, institutions, and luck acting together. Following those interactions can make achievement more intelligible, inequality more explainable, and possible changes more concrete—without pretending that explanation alone determines what a better outcome should be.

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